

Development of Blood Vessels

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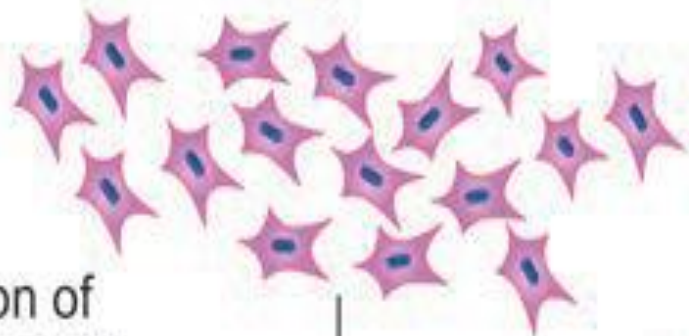
• **COMPONENTS OF BLOOD VASCULAR SYSTEM**

- Development of separate blood vascular system starts at the beginning of the 3rd week
- by formation of blood cells & blood vessels in the extraembryonic mesoderm of yolk sac, connecting stalk & chorion
- In the intraembryonic mesoderm, they develop 2 days later
- internal surfaces of blood vessels are lined by a layer of flattened cells called ***endothelium***
- endothelium is supported, on the outside, by varying amount of muscle & connective tissue
- components of blood vessels are of mesodermal origin (Tunica intima, media & externa)

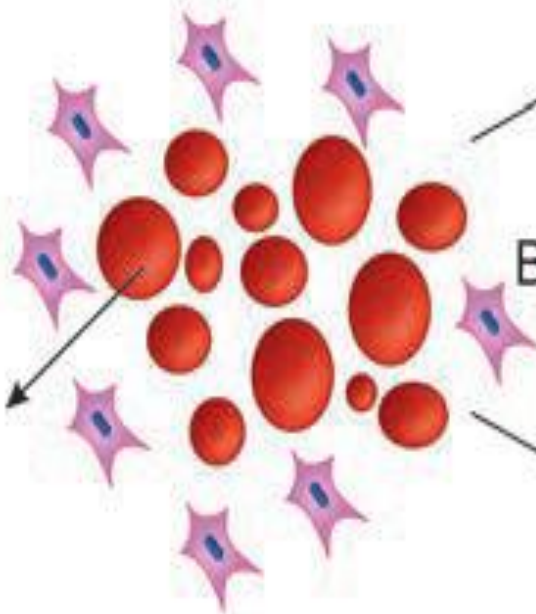
- first blood cells formed by induction of mesodermal cells to hemangioblasts (common precursor for blood cells and vessels)
- Hemopoiesis first occurs before birth in the yolk sac & allantois end of 3rd week
- later in the liver, spleen, thymus & lymph nodes of a foetus in the 5th week
- last 3 months before birth, red bone marrow becomes the primary site of hemopoiesis
- continues to be the source of blood cells after birth & throughout life
- blood vessels develop by 2 processes, i.e.
- **Vasculogenesis & angiogenesis**

- **Vasculogenesis**
- process of formation of new vessels by coalescence (assembly) of angioblasts (specialized mesenchymal cells)
- differentiation of angioblasts into endothelial cells & formation of primitive vascular network
- mesenchymal cells differentiate into angioblast cells
- Aggregations of angioblasts form the blood islands
- clefts appear in the blood islands
- clefts communicate with one another forming capillary plexus
- Reorganization of cells in islands leads to the formation of blood cells & blood vessels

Mesenchymal cells



Differentiation of mesenchymal cells into angioblast cells



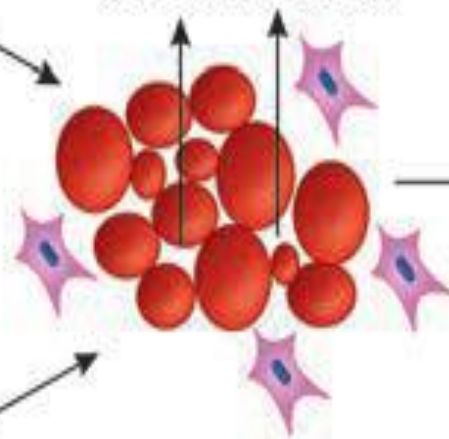
Angioblast cells

Aggregations of angioblasts form the blood islands

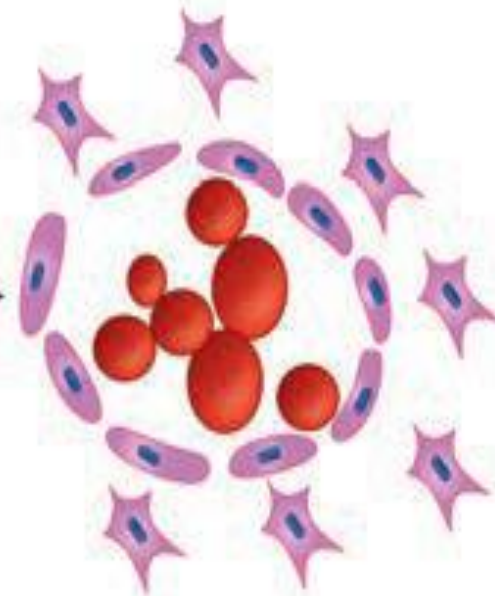


Blood islands

Clefts appear in the blood islands

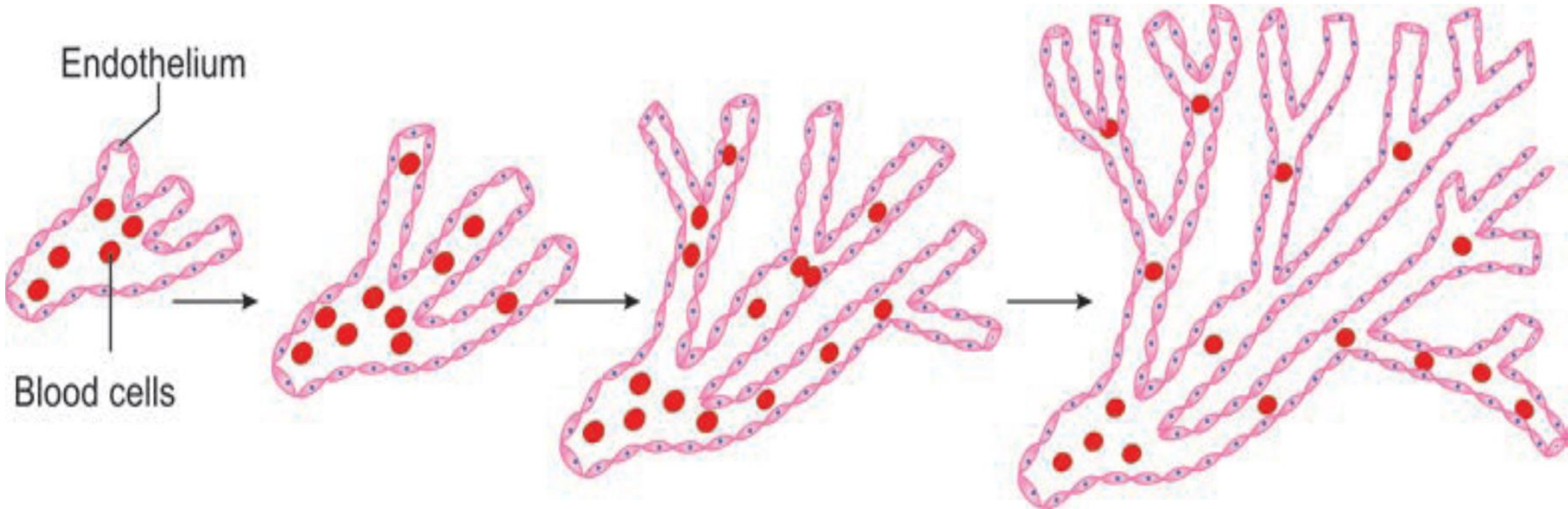


Reorganization of cells in islands to form blood cells and blood vessels



- **Angiogenesis**

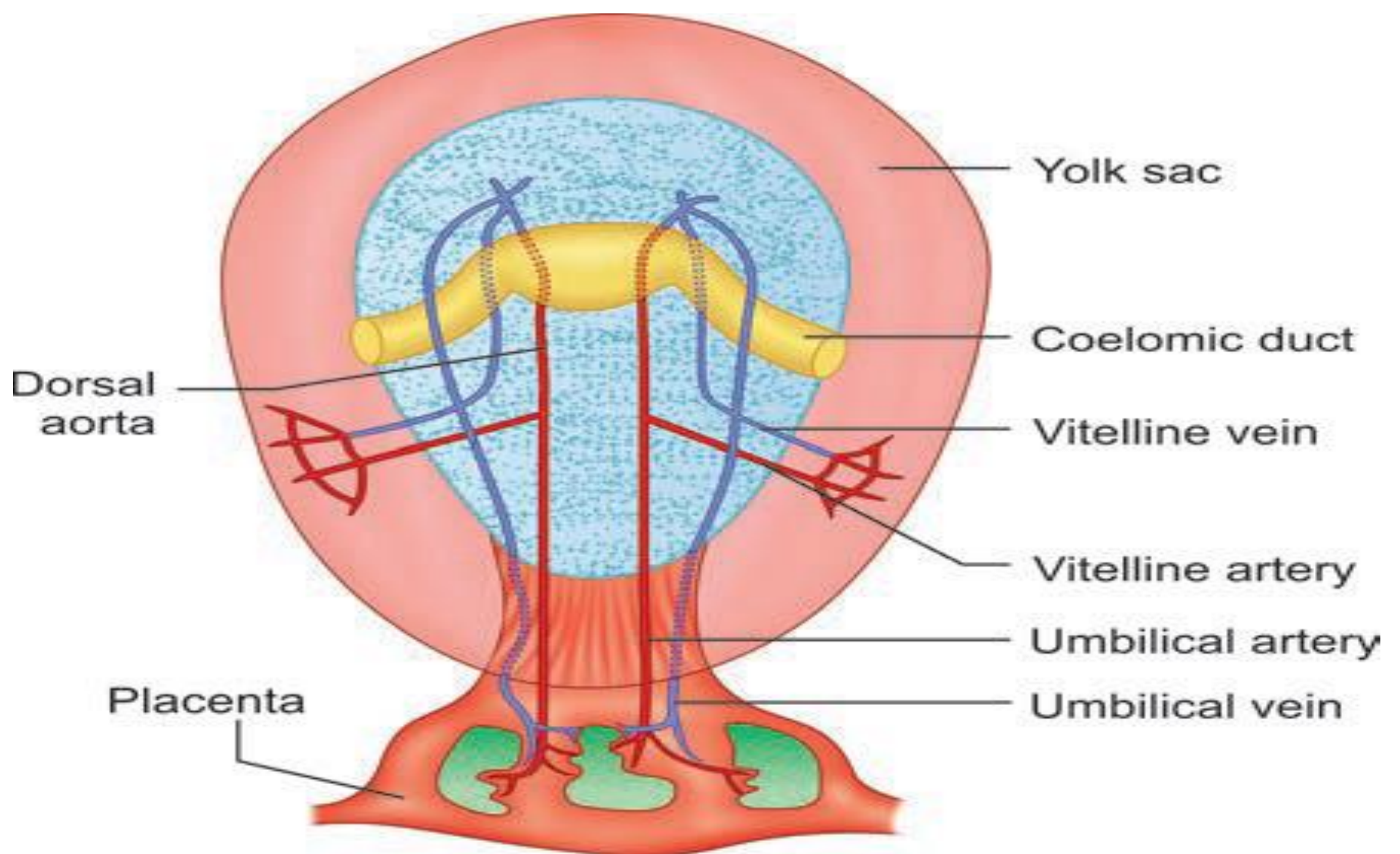
- process of sprouting of new vessels from existing ones
- Mesenchyme differentiates, over the yolk sac, in the connecting stalk, in the body of the embryo form small masses of *angioblastic tissue*
- angioblastic tissue gives rise to *endothelium* and also to *blood cells*
- first blood vessels are derived from this endothelium



- vessels proliferate in number and become interconnected to form a vascular system
- primitive heart begins to pump blood through this network of vessels
- Outcome is that nutrition from the placenta & yolk sac can be made available to the growing embryo
- Angiogenesis begins first in the yolk sac wall during 3rd week
- Inside the embryo it begins in 5th week

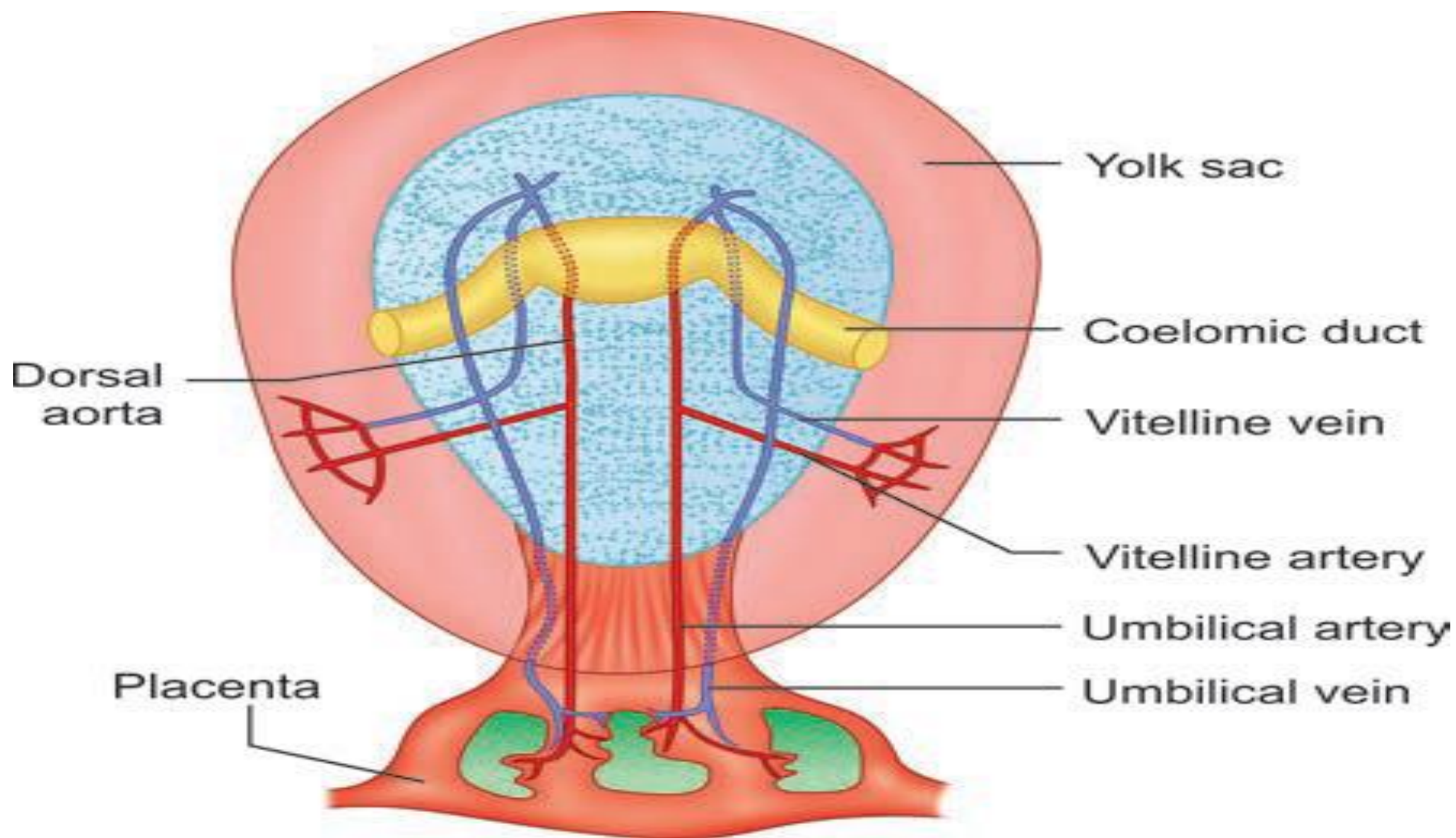
- Molecular regularization of vasculogenesis is by fibroblast growth factor 2 (FGF2) and vasoendothelial growth factor (VEGF)
- Sprouting of new vessels from existing one is by VEGF
- maturation and modeling of vasculature is by platelet-derived growth factor (PDGF) and transforming growth factor-B (TGF-B

- **EXTRAEMBRYONIC BLOOD VASCULAR SYSTEM**
- 3rd week of development, the extraembryonic blood cells & blood vessels develop in the following regions:-
 - Wall of yolk sac
 - Connecting stalk
 - Chorion
- Those developing in the :-
 - wall of yolk sac become the ***vitelline vessels***
 - connecting stalk & chorion become the ***umbilical vessels***

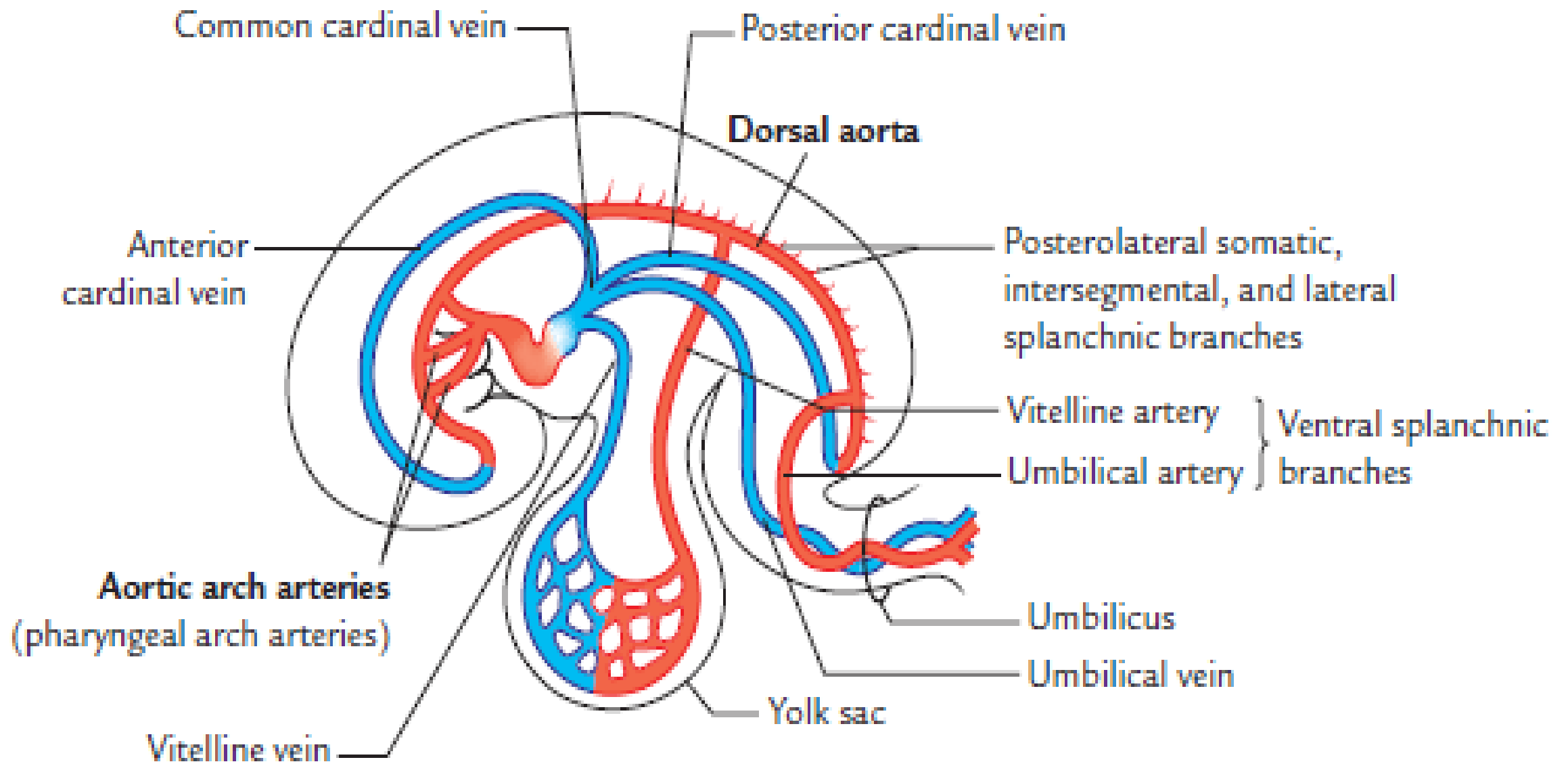


- **INTRAEMBRYONIC BLOOD VASCULAR SYSTEM**

- later part of 3rd week of development
- the intraembryonic blood cells and blood vessels are formed in the intraembryonic mesoderm and establish connection with the extraembryonic vessels
- Before the formation of embryonic folds, 2 longitudinal vessels (*dorsal aortae*) develop in the flattened embryonic disc
- are on either side of notochord and along the dorsal wall of yolk sac



- dorsal aortae at the cephalic end of embryonic disc invade the cardiogenic area & join the primitive heart tubes
- At the caudal end of the embryonic disc, the dorsal aortae extend into the connecting stalk as ***umbilical arteries***
- Some blood vessels sprout from each dorsal aorta into the yolk sac that forms the ***vitelline arteries***
- ***umbilical veins*** develop in relation to somatopleuric layer of intraembryonic coelom
- ***vitelline veins*** develop in the splanchnopleuric layer of intraembryonic coelom
- ***Cardinal veins*** develop in the body wall of embryo



2 Vessels as seen in a 26-day-old embryo.

- Umbilical and vitelline veins pass through the septum transversum & join the cranial end of each primitive heart tube
- Cardinal veins also join the cranial ends of primitive heart tubes

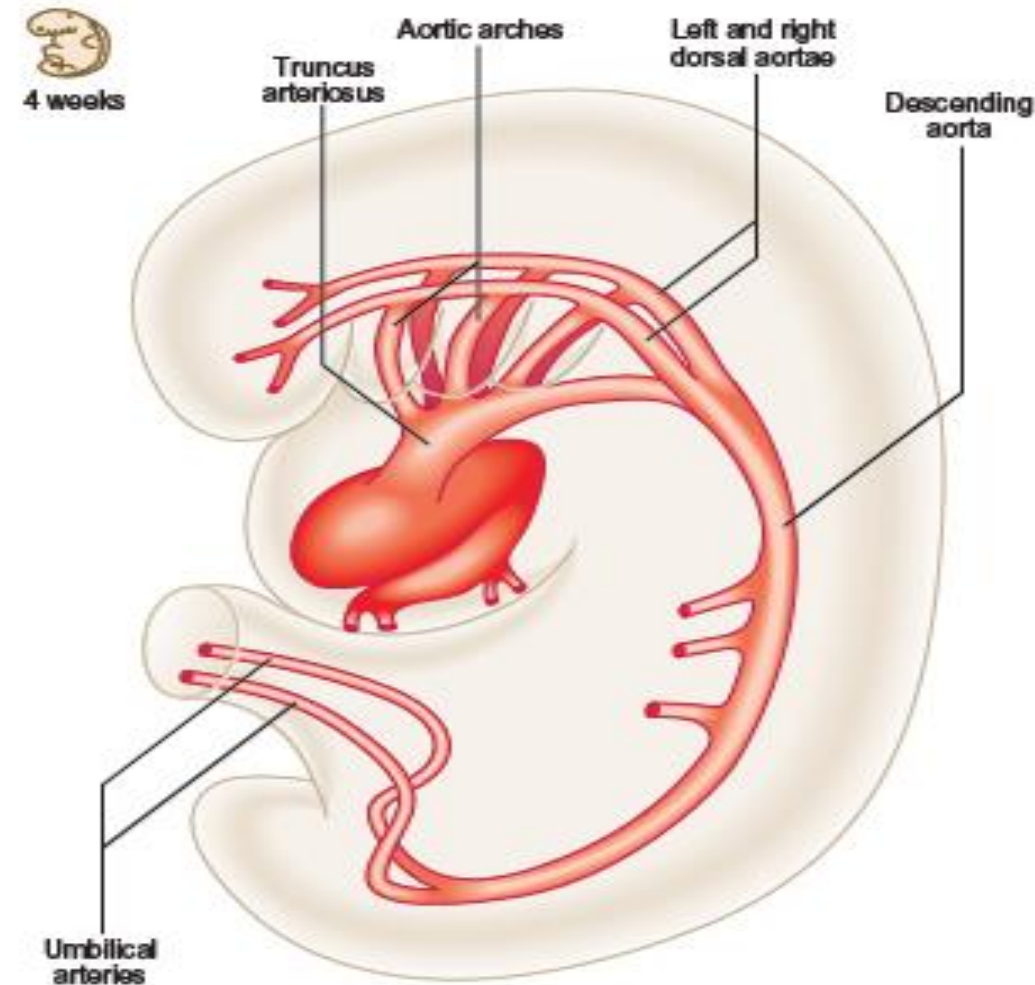


Fig. 6.10 The arterial system in a 4-week-old embryo.

- **Development of Arterial System**
- arteries of body develop from 2 main sources:
- **pharyngeal (aortic) arch arteries & dorsal aorta**
- *aortic arches* give rise to the arteries of head and neck region
- *dorsal aorta* gives rise to the arteries of rest of the body

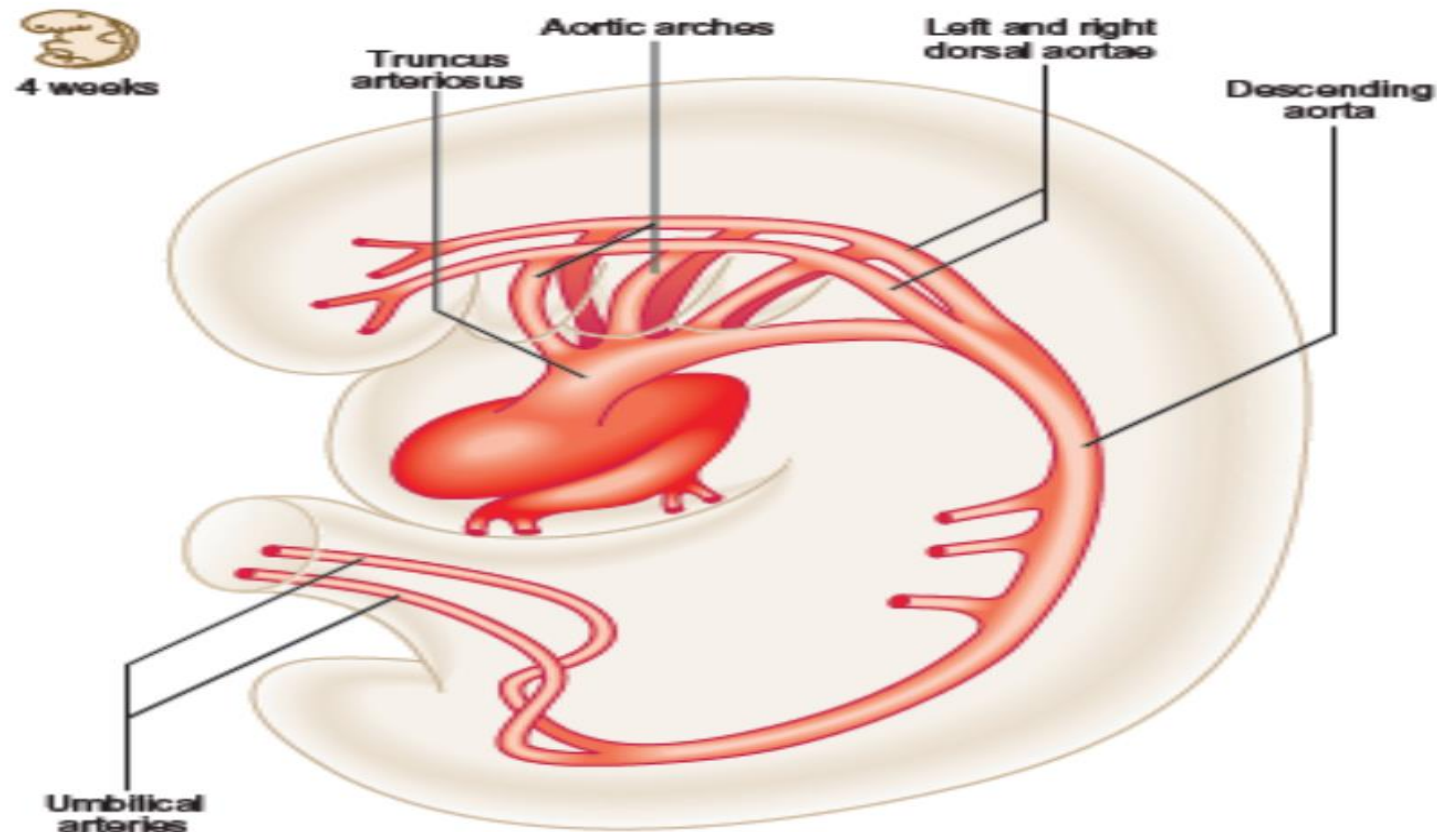


Fig. 6.10 The arterial system in a 4-week-old embryo.

- **Pharyngeal Arch Arteries and Their Derivatives**
- First arteries to appear in a developing embryo are **right and left primitive aortae**
- become continuous with 2 primitive heart tubes
- Each primitive aorta is divided into 3 parts:
- ventral aorta, first aortic arch artery & **dorsal aorta** as follows
- **(a)** part lying ventral to foregut is called *ventral aorta*
- **(b)** part lying dorsal to foregut is called *dorsal aorta*
- **(c)** arched portion connecting the ventral & dorsal aortae lies in the first pharyngeal arch
- It forms *first aortic arch artery*

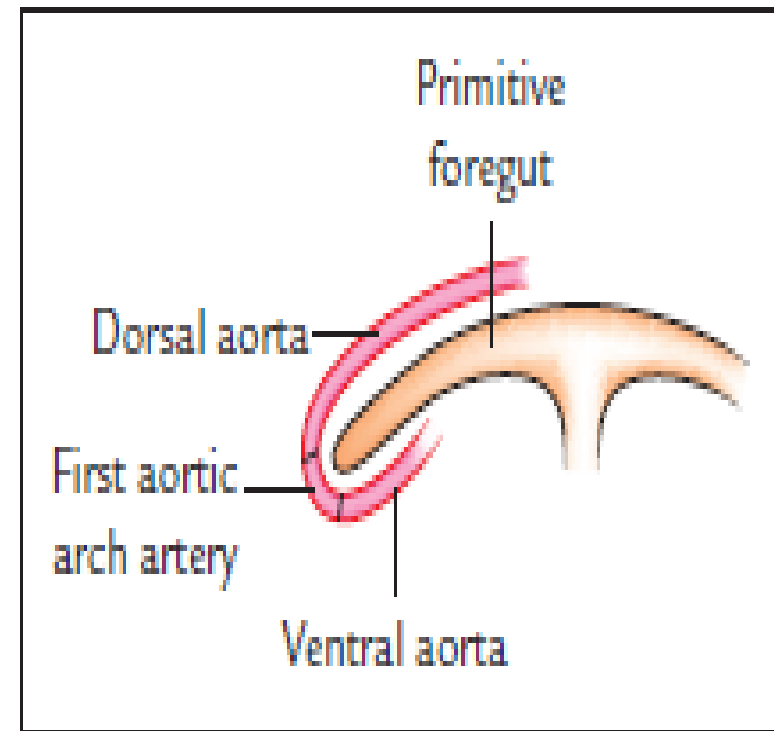
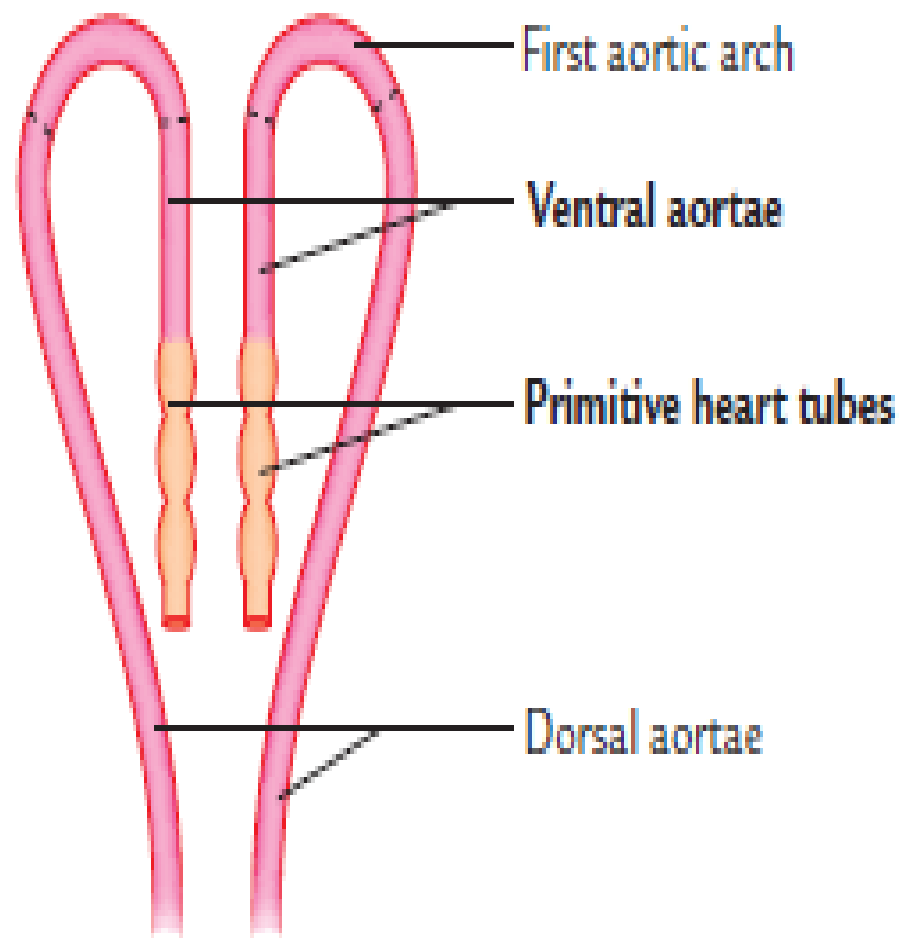
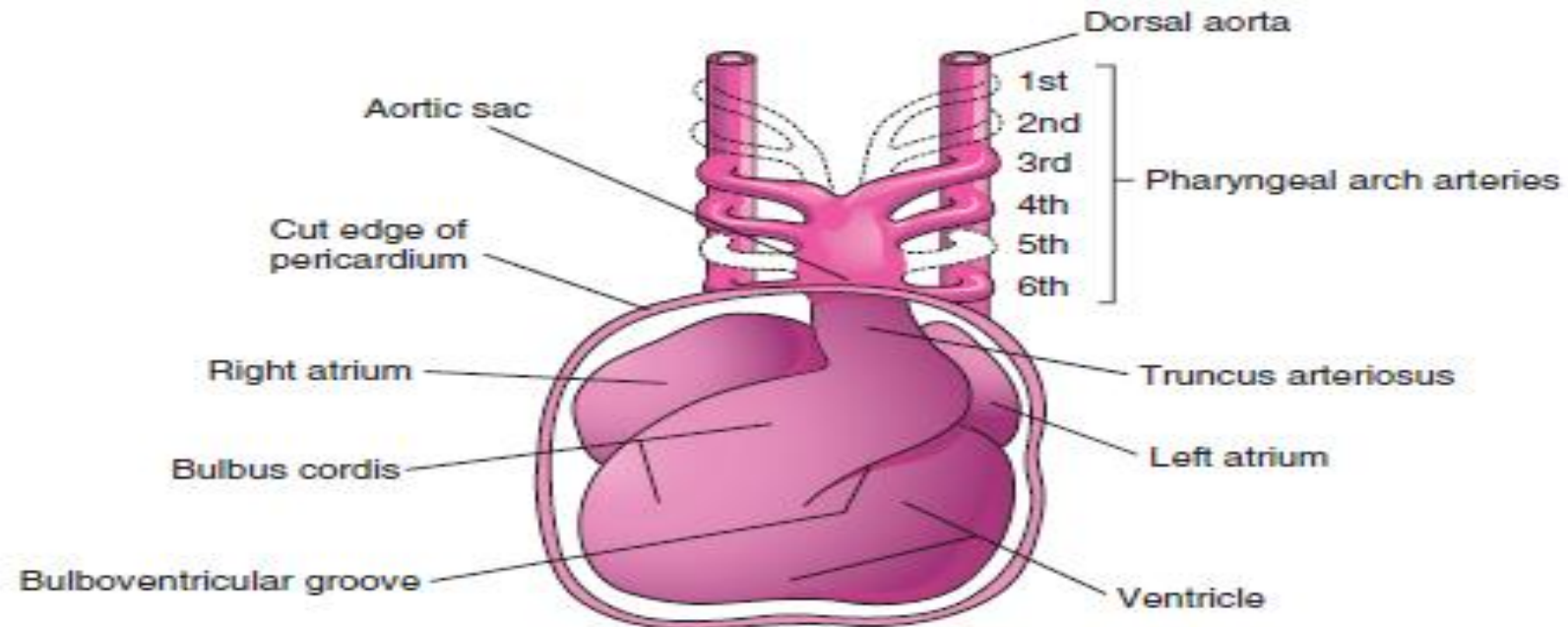


Fig. 19.3 Formation of primitive aorta. Figure in the inset shows subdivisions of the primitive aorta.

- After fusion of 2 primitive heart tubes the 2 ventral aortae partially fuse to form ***aortic sac***
- unfused parts remaining as the ***right & left horns*** of the sac
- pharyngeal arches develop during 4th & 5th weeks, each arch is supplied by an artery—the **pharyngeal arch artery** derived from the aortic sac
- aortic sac gives rise to 6 pairs of aortic arch arteries
- aortic arch arteries are embedded in the mesenchyme of the pharyngeal arches
- each being connected ventrally to the right or left horn of the aortic sac & dorsally to the dorsal aorta



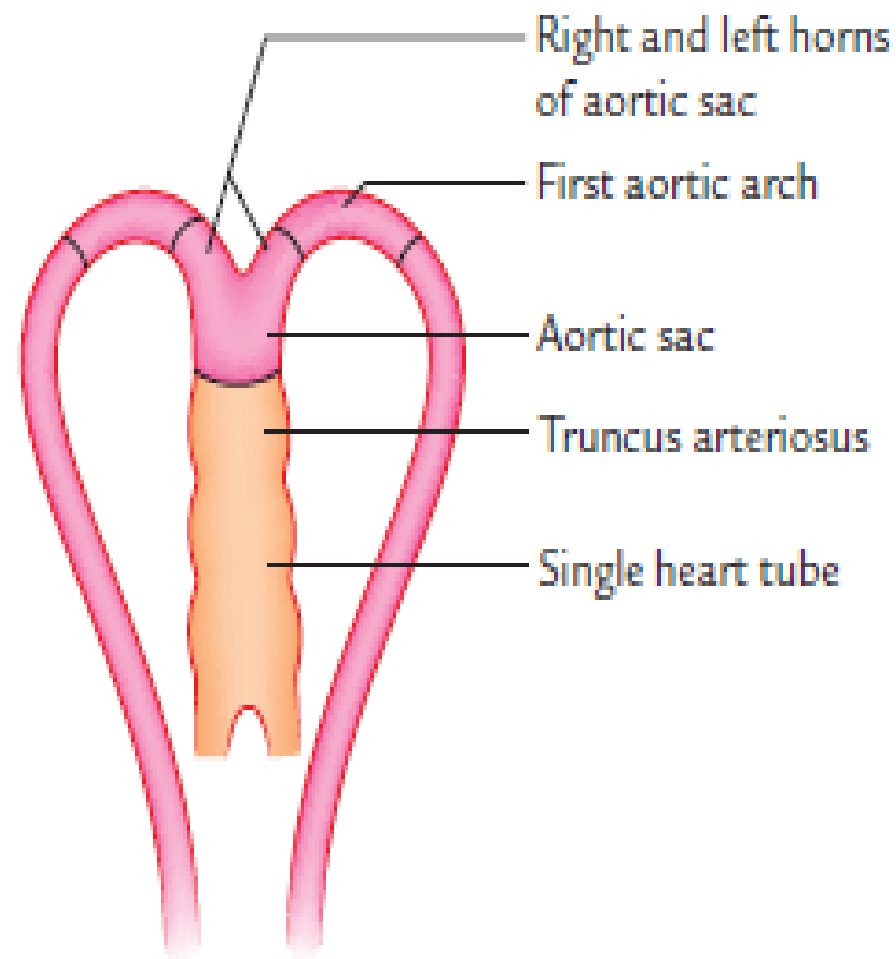


Fig. 19.4 Formation of aortic sac and its fusion with single heart tube.

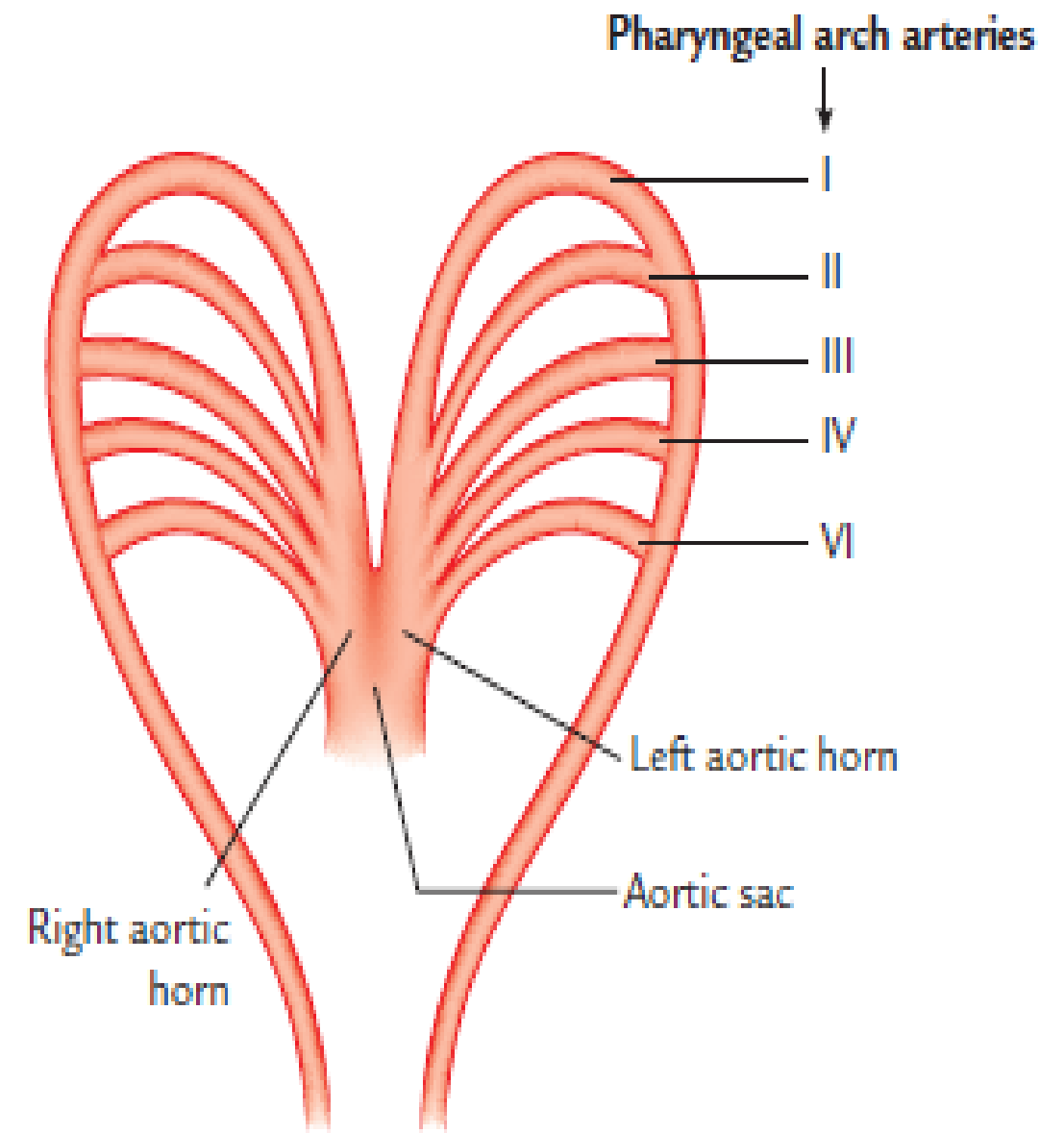
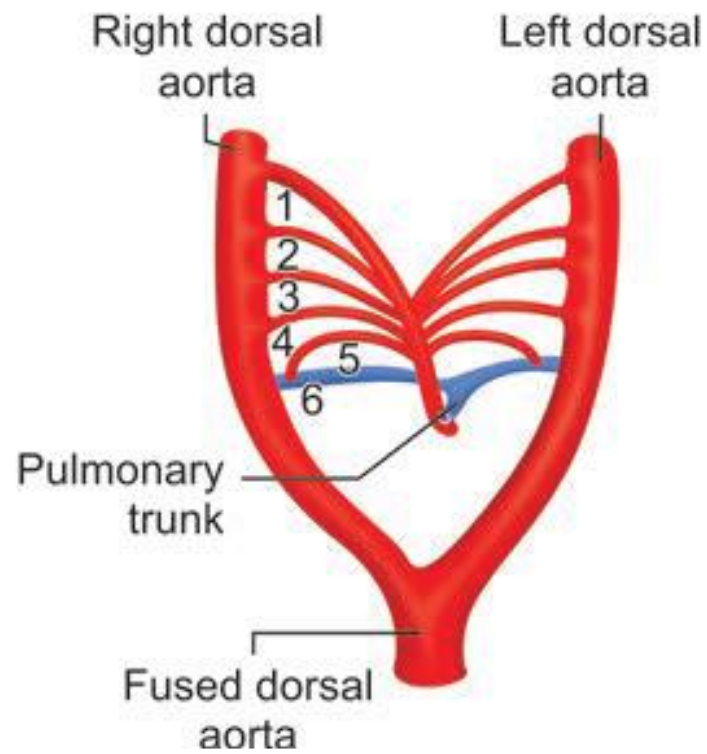
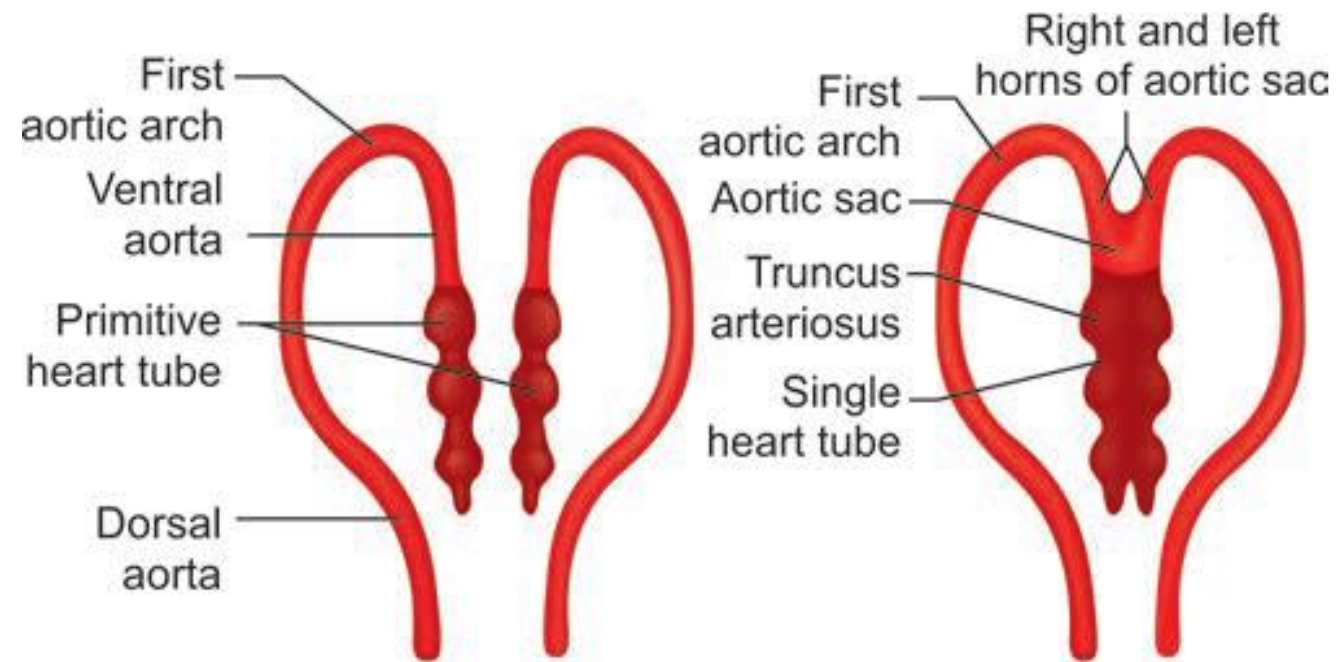
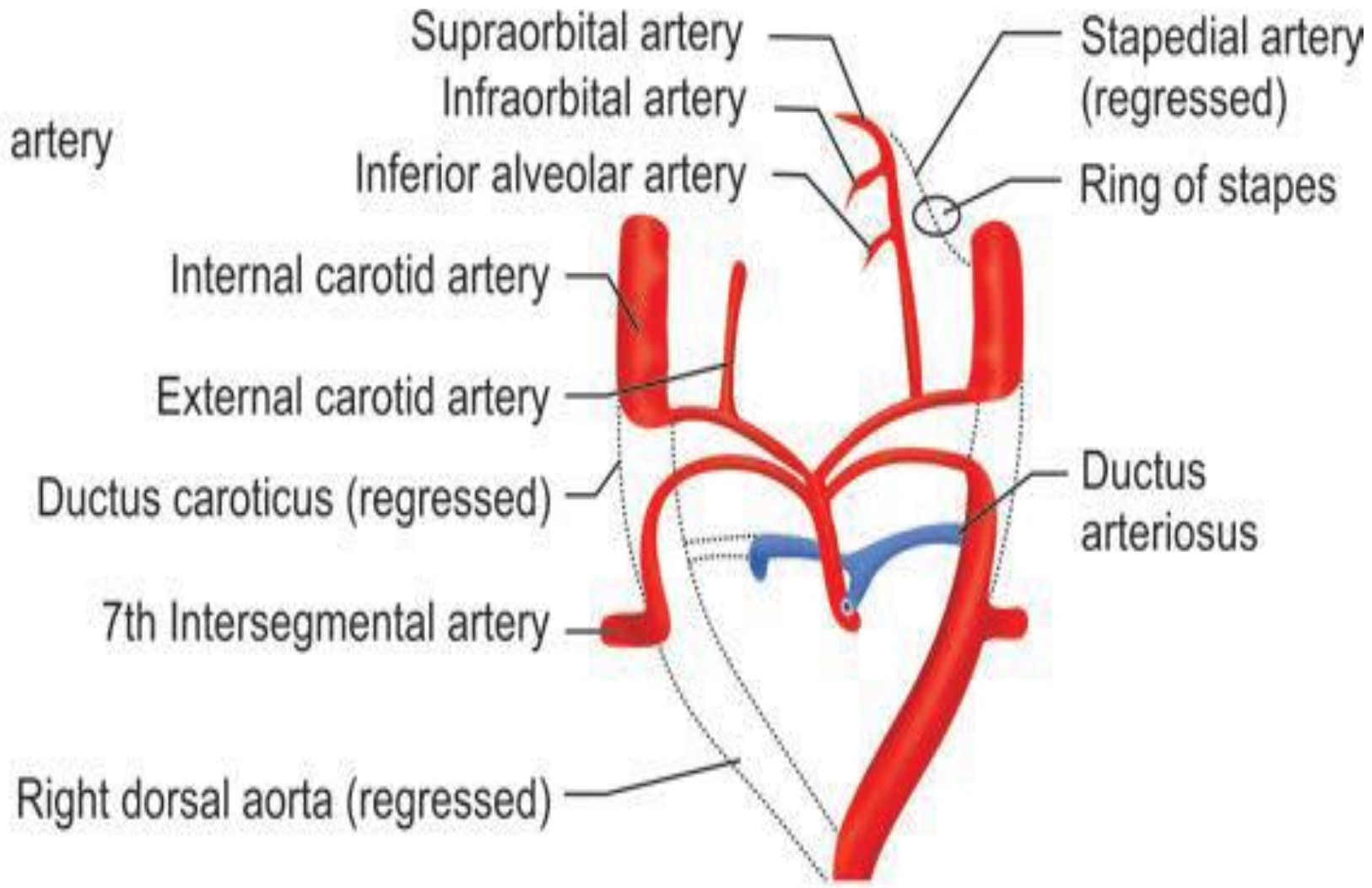
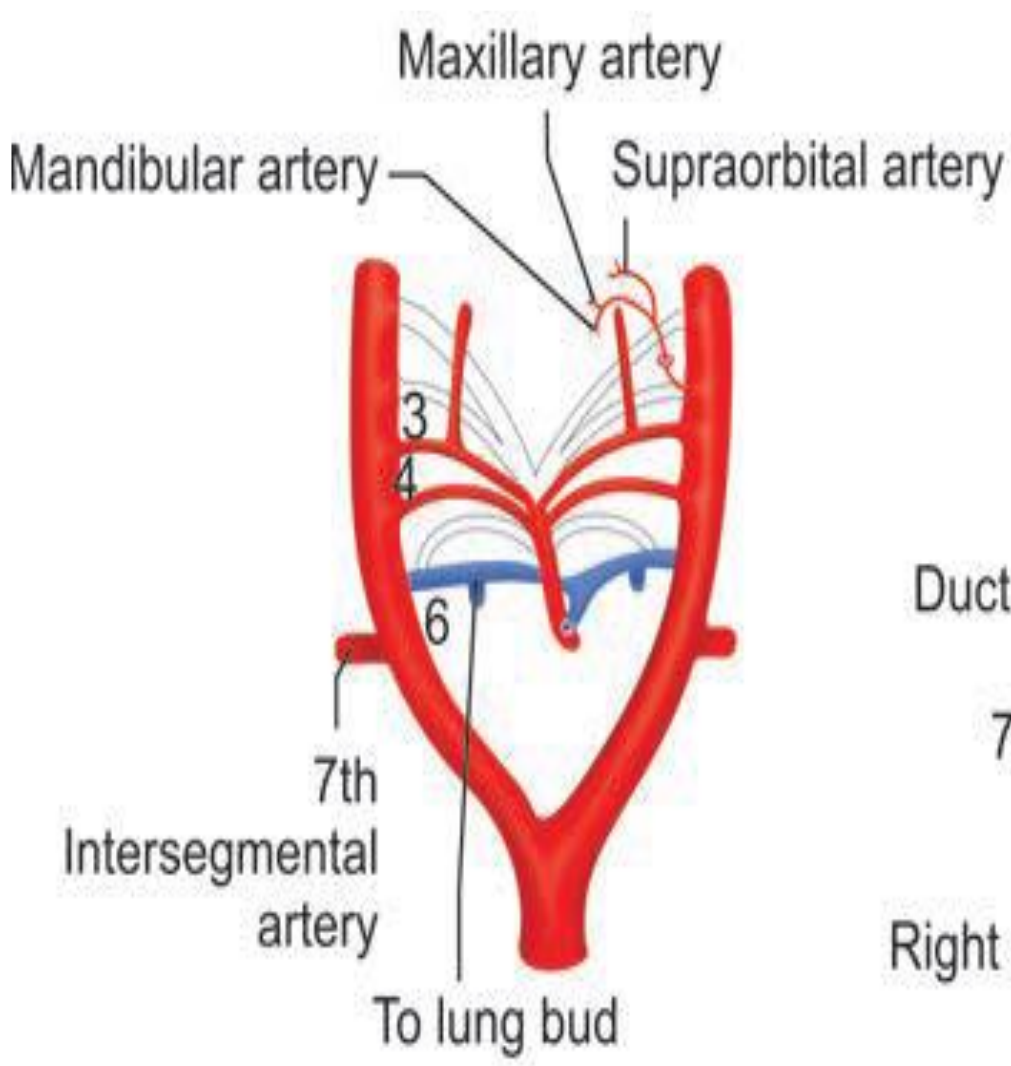


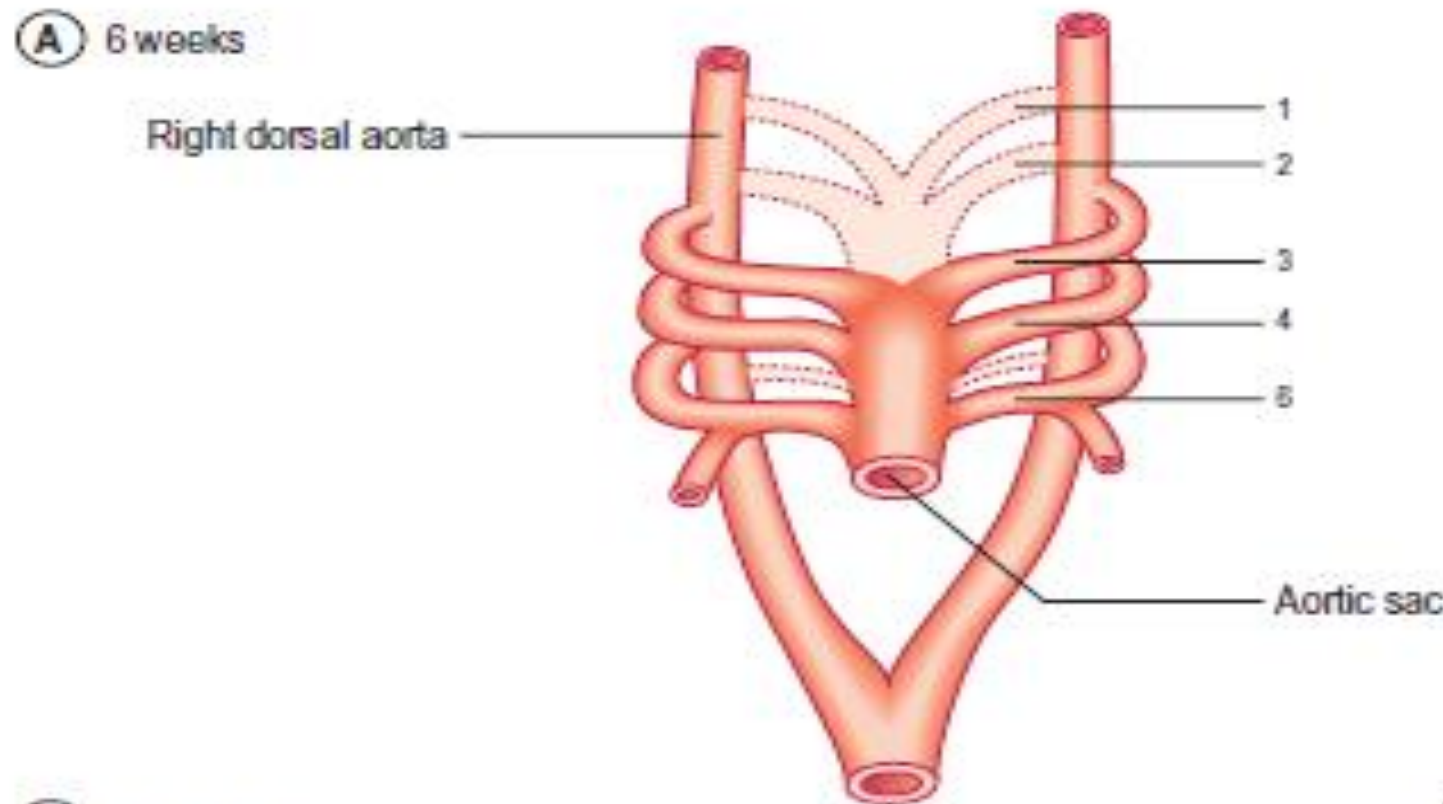
Fig. 19.5 Aortic arches (pharyngeal arch arteries).



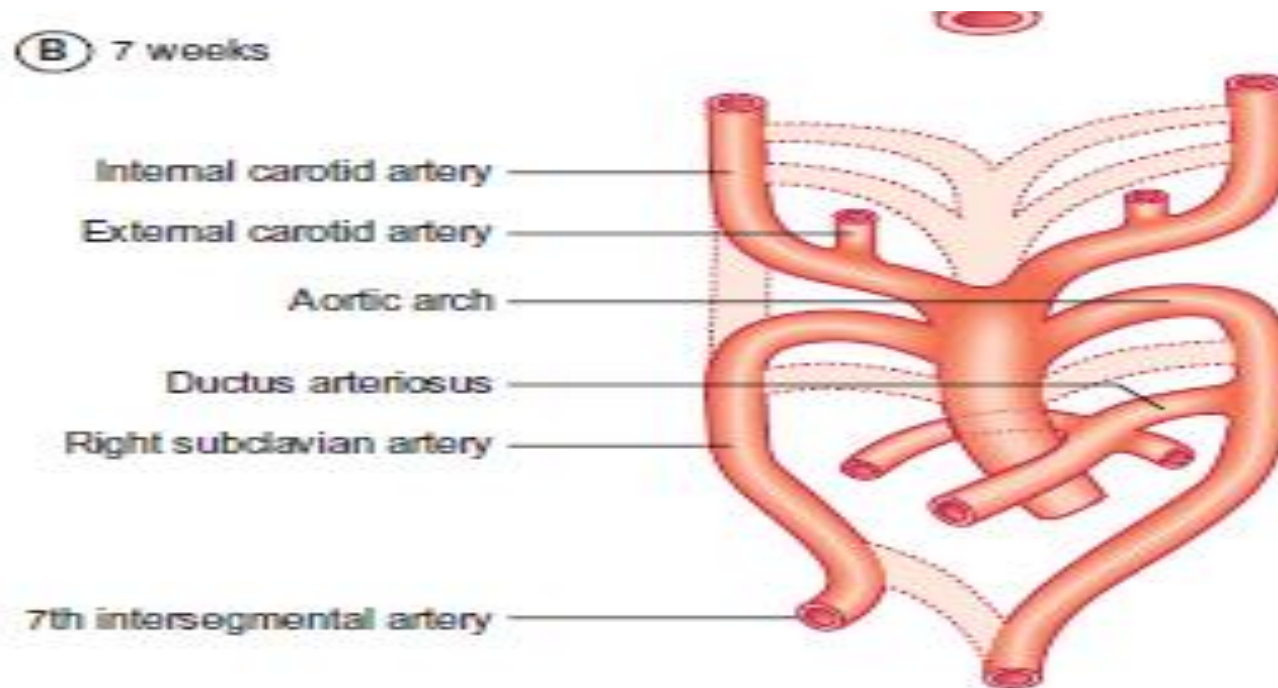
- major arteries of the head & neck, and of the thorax, are derived from these arches as follows:
- greater part of the 1st & 2nd arch arteries disappear
- In adult life, the 1st arch artery is represented by the ***maxillary artery***
- 2nd arch artery persists for some part of fetal life as the ***stapedial artery***
- 5th arch artery also disappears
- aortic sac is connected only with the arteries of the 3rd, 4th and 6th arches

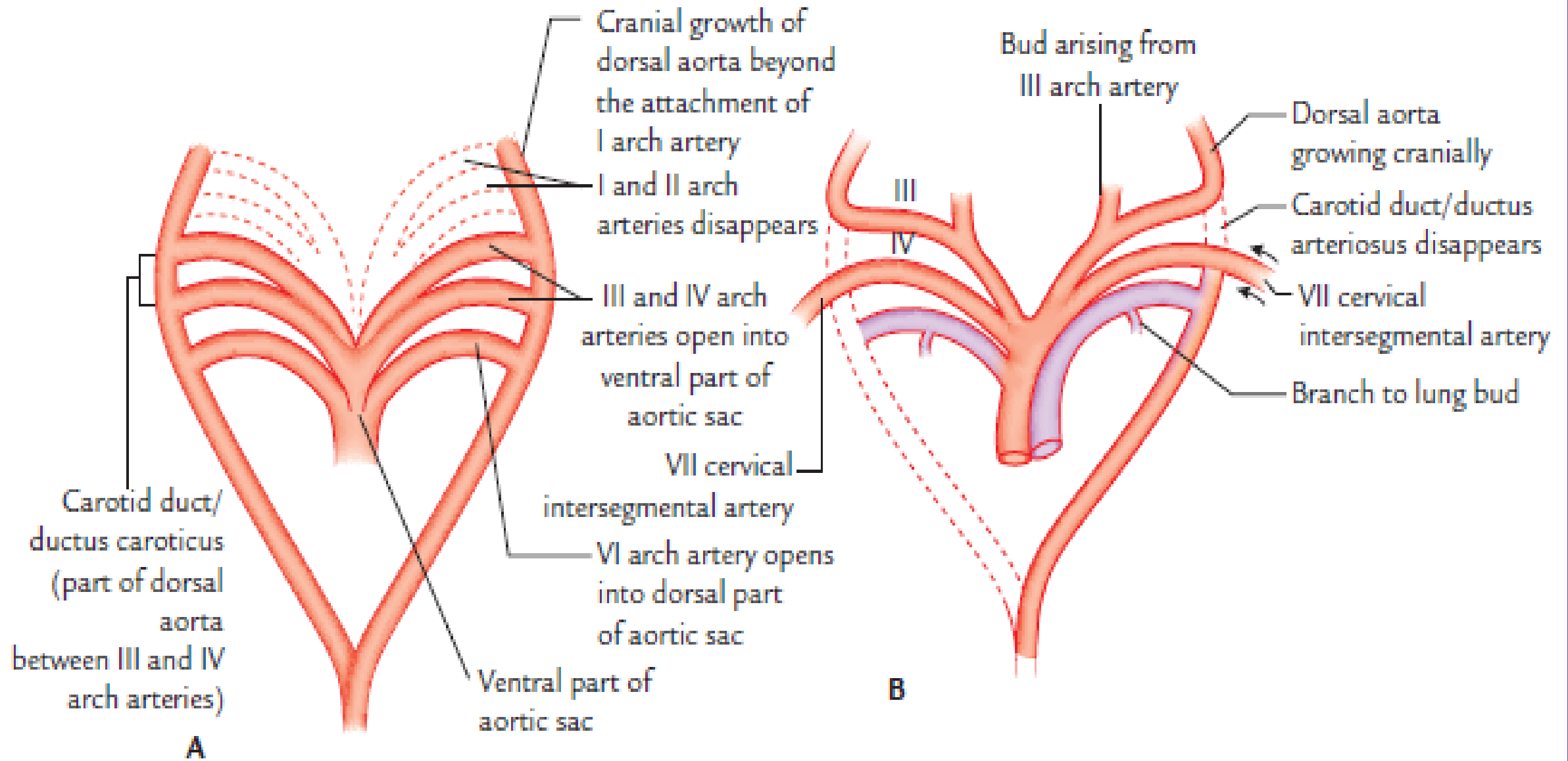


- **3rd & 4th arch arteries** open into the **ventral part** & **6th arch artery** into the **dorsal part**, of the aortic sac
- spiral septum that is formed in the truncus arteriosus extends into the aortic sac
- Fuses with its posterior wall in such a way that blood from the pulmonary trunk passes only into the 6th arch artery
- while that from the ascending aorta passes into the 3rd & 4th arch arteries



- Several changes take place in the arterial arches to produce the adult pattern as follows:
- 2 dorsal aortae grow cranially, beyond the point of attachment of the 1st arch artery
- portion of the dorsal aorta, between the attachment of the 3rd & 4th arch arteries (***ductus caroticus***), disappears on both sides
- portion of the right dorsal aorta, between the point of attachment of the 4th arch artery & the point of fusion of the 2 dorsal aortae, disappears





- Each 6th arch artery gives off an artery to the developing lung bud
- On the right side, the portion of the 6th arch artery between this bud & the dorsal aorta disappears
- On the left side, this part remains patent & forms ***ductus arteriosus***
- Ductus arteriosus carries most of the blood from the right ventricle to the dorsal aorta
- It is obliterated after birth & is then seen as the ***ligamentum arteriosum***
- Each 3rd arch artery gives off a bud that grows cranially to form the ***external carotid artery***

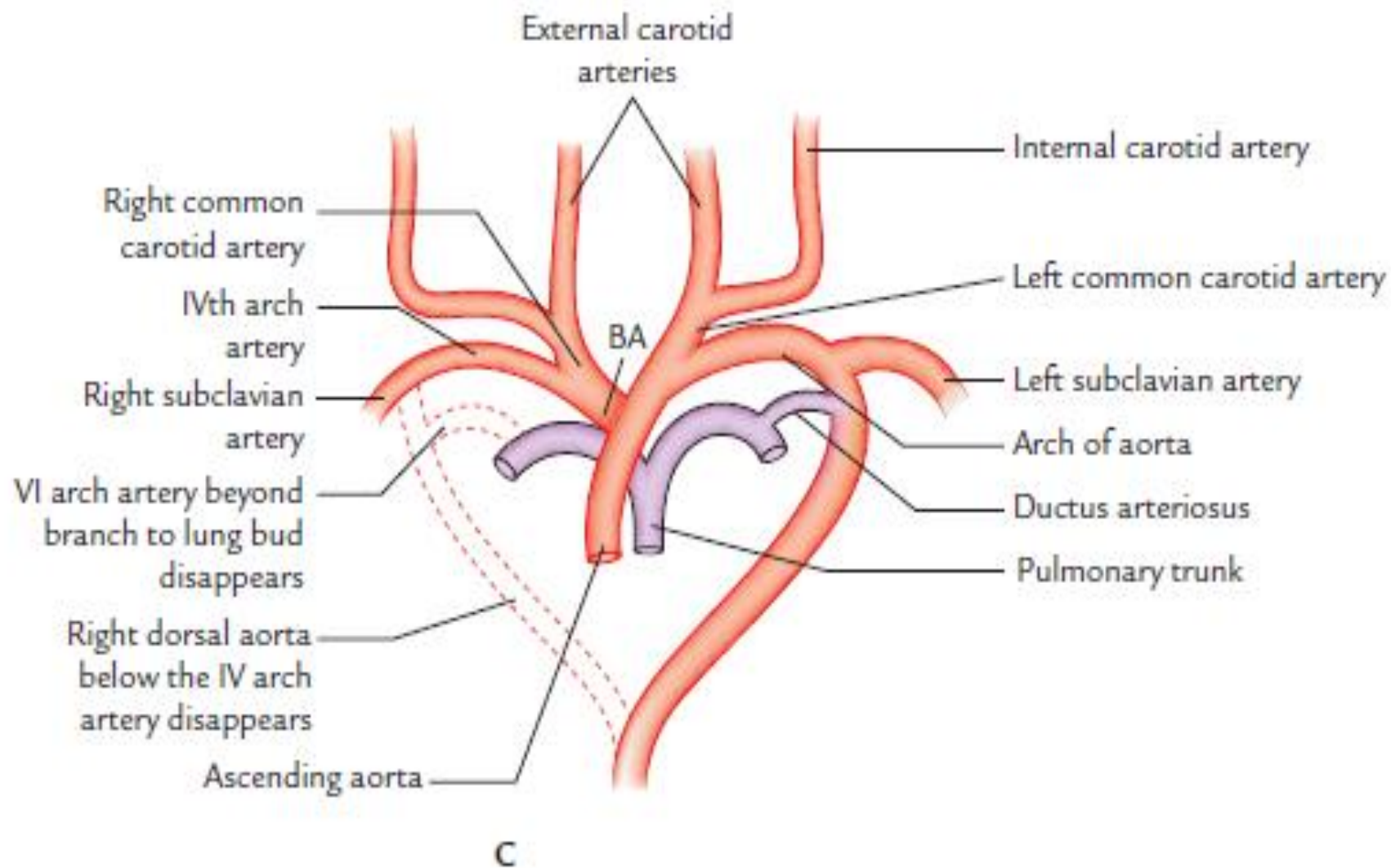
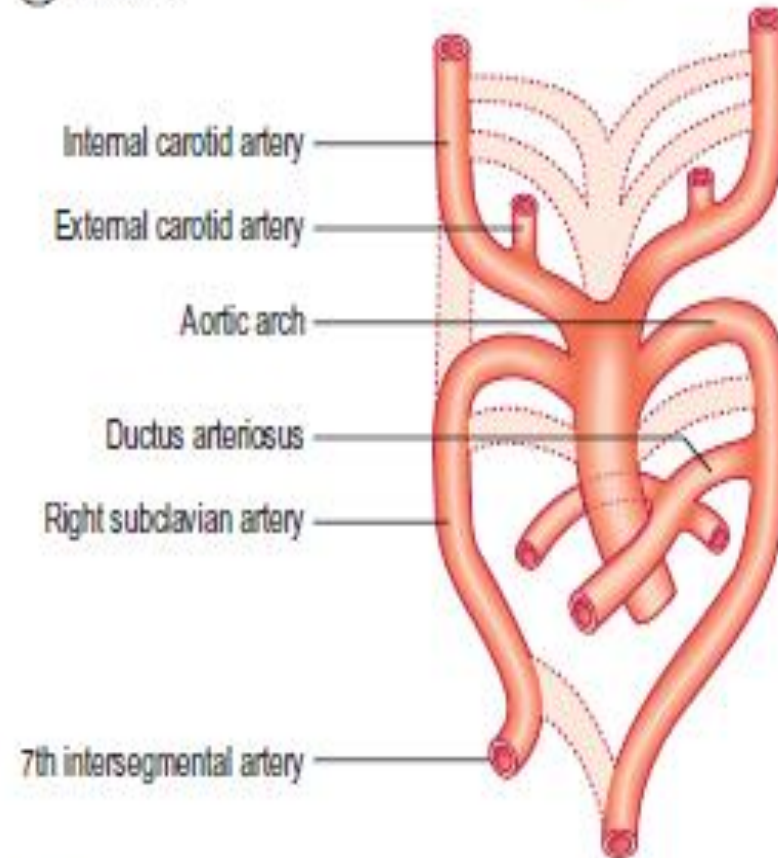


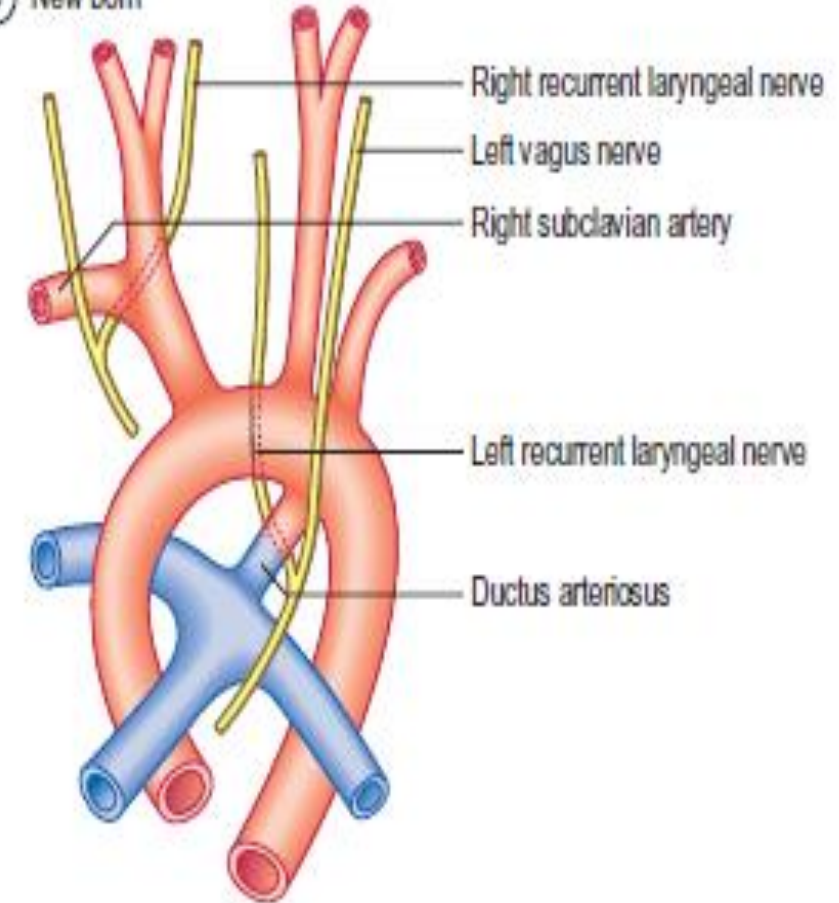
Fig. 19.6 Development of main arteries of head, neck, and thorax. BA = brachiocephalic artery.

- dorsal aortae give off a series of lateral intersegmental branches to the body wall
- the **seventh cervical intersegmental artery** supplies the upper limb bud
- It comes to be attached to the dorsal aorta near the attachment of the 4th arch artery

(B) 7 weeks



(C) New born



6.11 The aortic arches. (A) 6 weeks; (B) 7 weeks; (C) newborn.

- The development of the main arteries can now be summarized as follows:
- ***ascending aorta & pulmonary trunk*** are formed from the truncus arteriosus
- ***arch of the aorta*** is derived from the ventral part of the aortic sac, its left horn, & the left 4th arch artery
- ***descending aorta*** is derived from the left dorsal aorta, below the attachment of 4th arch artery, along with the fused median vessel
- ***brachiocephalic artery*** is formed by the right horn of the aortic sac
- proximal part of the ***right subclavian artery*** is derived from the right 4th arch artery
- remaining part of the artery being derived from the **seventh cervical intersegmental artery**

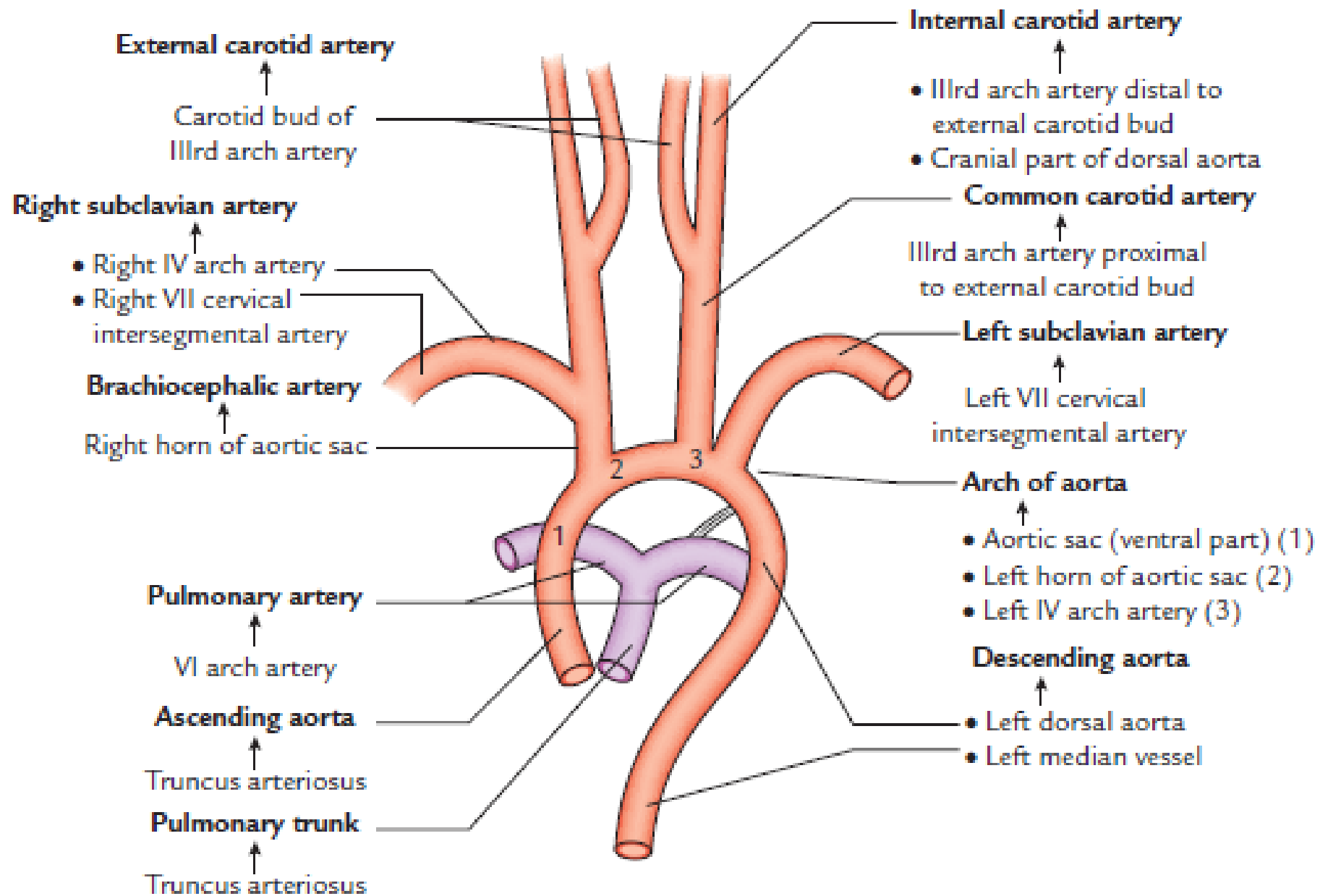


Fig. 19.7 Source of development of major arteries of head and neck, upper limb, and thorax.

- ***Left subclavian artery*** is derived from the seventh cervical intersegmental artery, arises from the dorsal aorta opposite the attachment of the 4th arch artery
- ***common carotid artery*** is derived, on either side, from part of the 3rd arch artery, **proximal to the external carotid bud**
- ***internal carotid artery*** is formed by the portion of the 3rd arch artery distal to the bud, along with the original dorsal aorta cranial to the attachment of the 3rd arch artery

- the right 3rd & 4th arch arteries arise from the right horn of the aortic sac, the common carotid & subclavian arteries become branches of the brachiocephalic artery
- **pulmonary arteries** are derived from the part of the 6th arch arteries lying between the pulmonary trunk & the branches to the lung buds

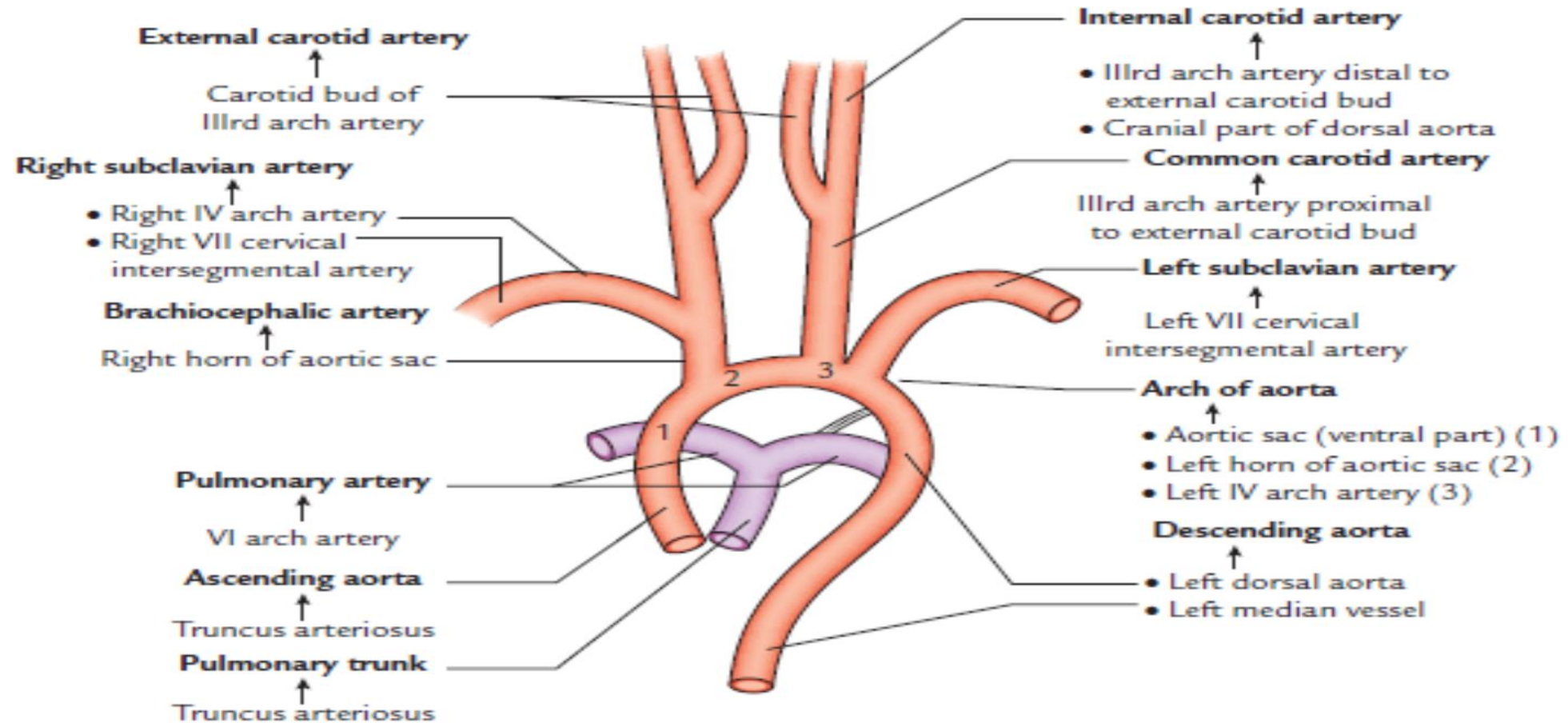


Fig. 19.7 Source of development of major arteries of head and neck, upper limb, and thorax.

- **DEVELOPMENT OF OTHER ARTERIES**

- primitive dorsal aortae give off three groups of branches
- These are as follows:
- **ventral splanchnic arteries** supply the gut
- Most of these arteries disappear but three arteries, the ***celiac, superior mesenteric and inferior mesenteric*** remain
- They supply the infradiaphragmatic part of the foregut, the midgut, and the hindgut respectively
- Other remnants of these vessels are the ***bronchial and esophageal*** arteries

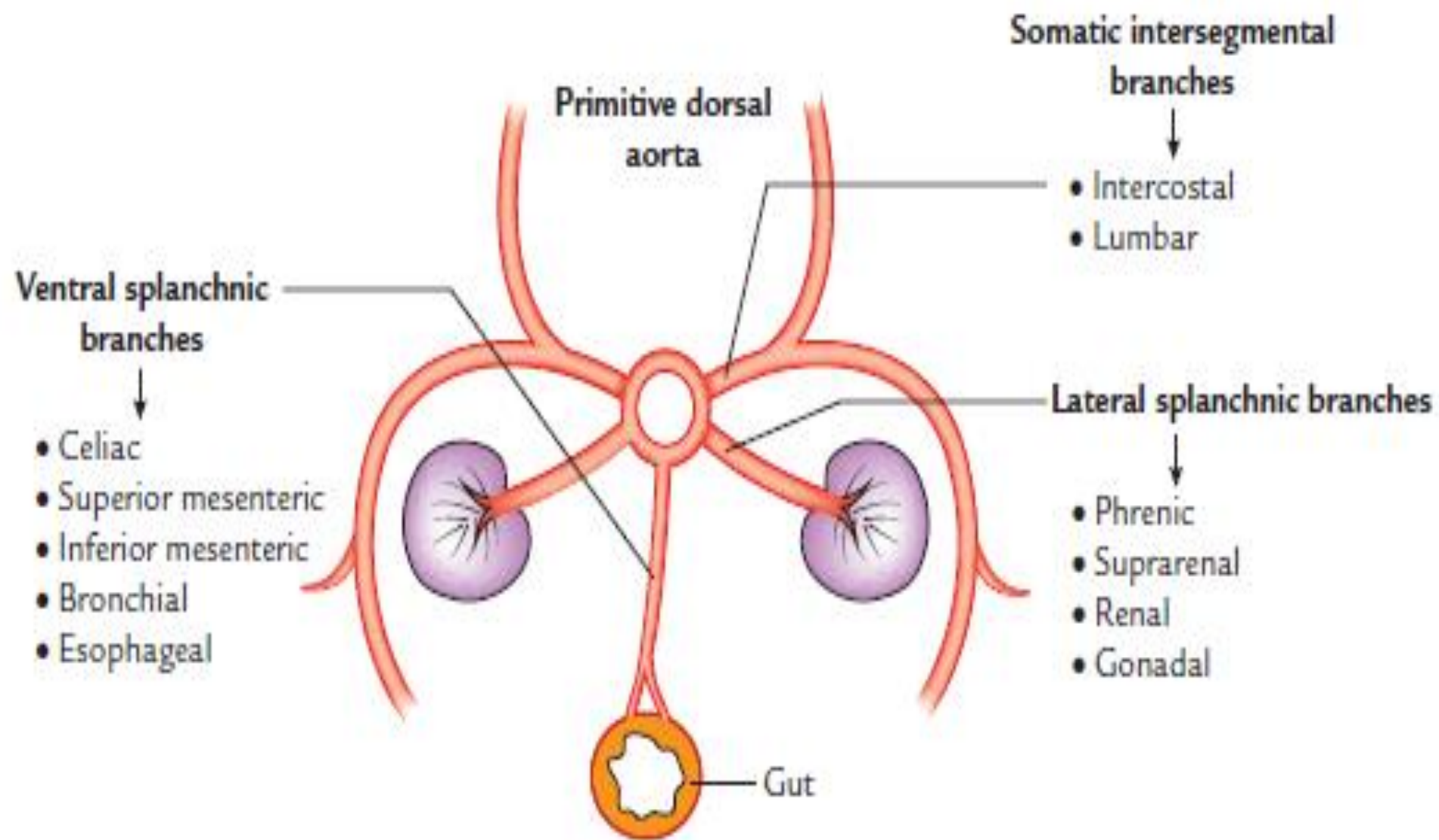
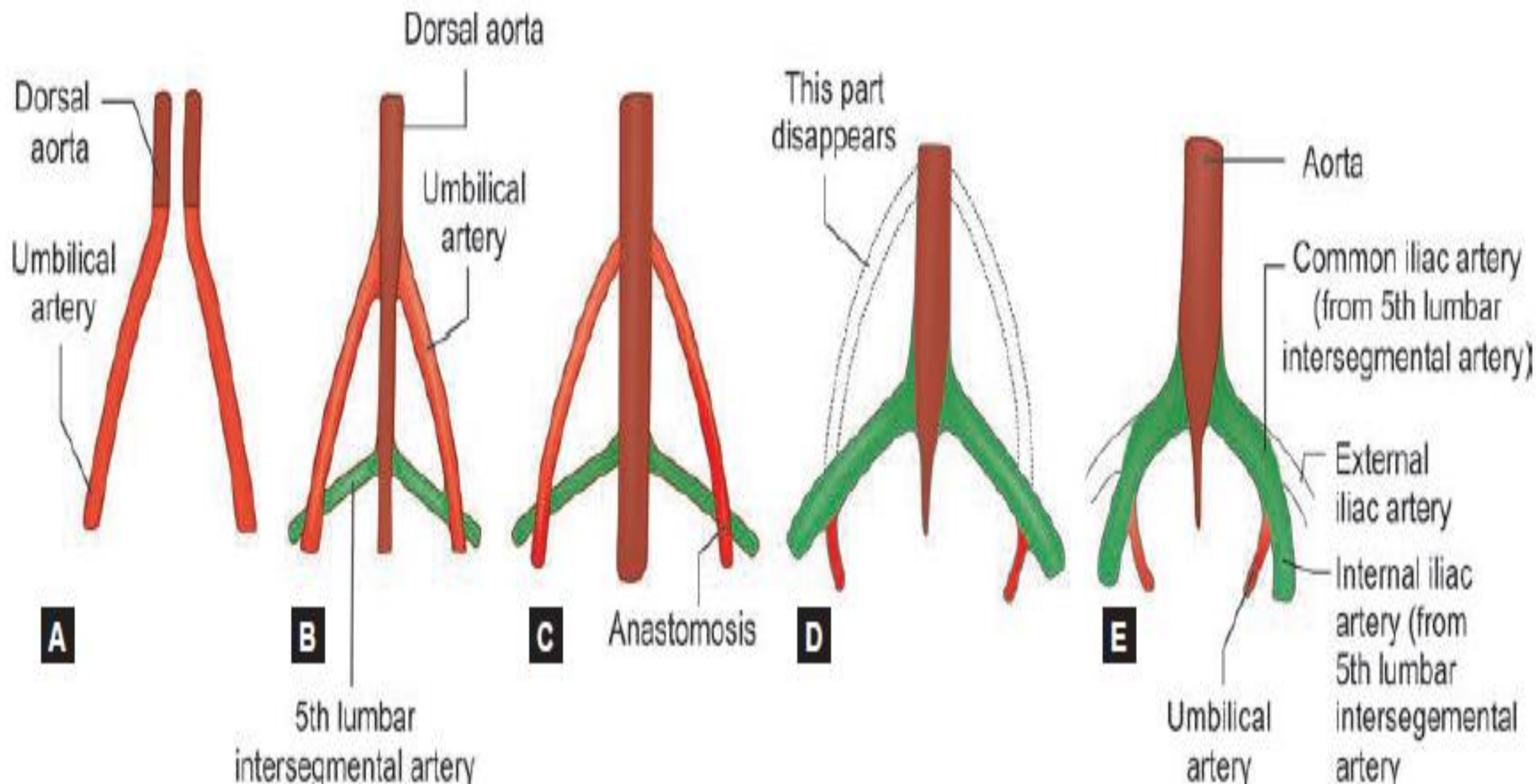


Fig. 19.12 Main branches of embryonic dorsal aorta.

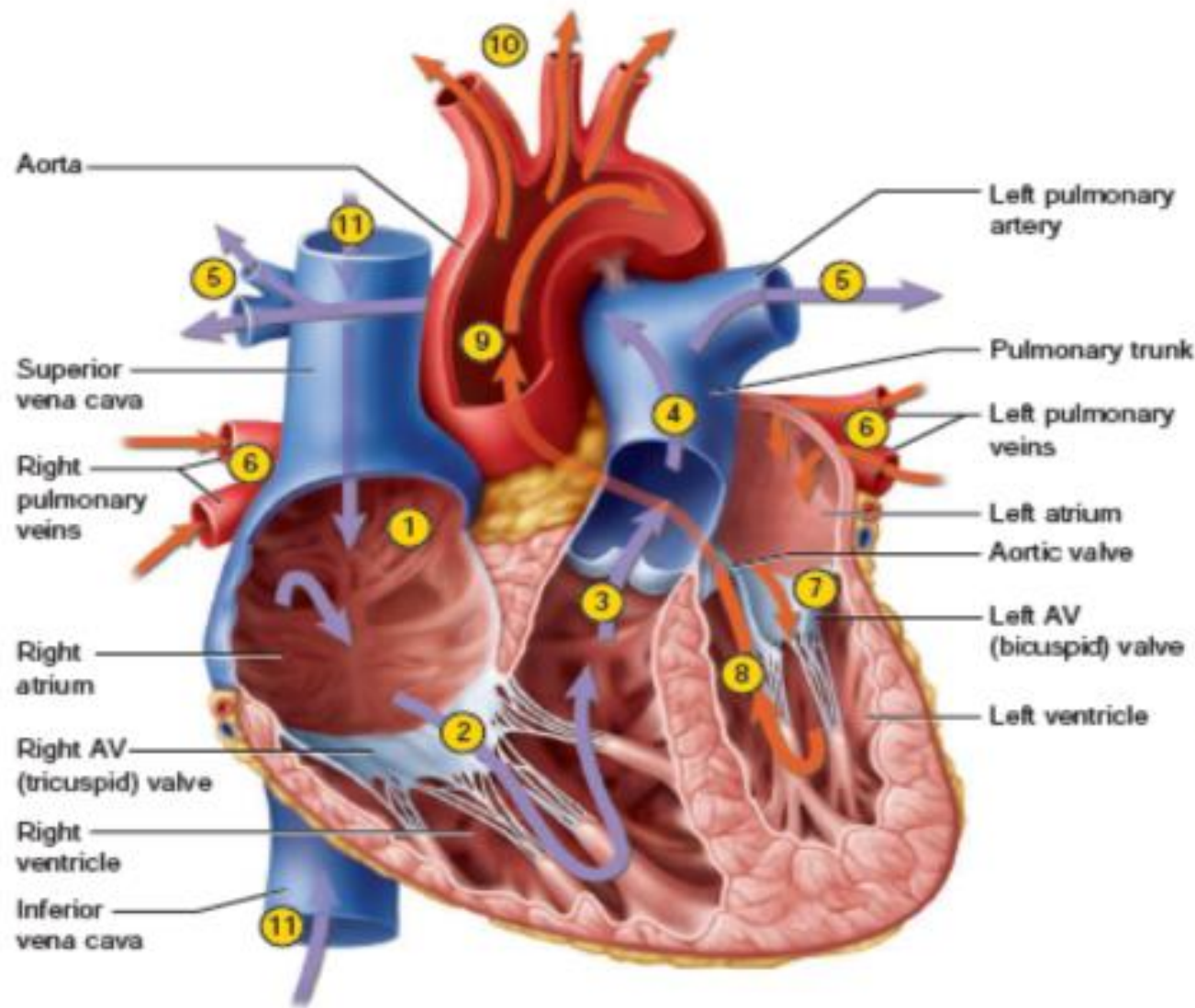
- ***lateral or intermediate splanchnic arteries*** supply structures developing from the **intermediate mesoderm**
- These persist as the *renal, suprarenal, phrenic, and spermatic or ovarian arteries*
- ***dorsolateral (somatic intersegmental)*** branches run between two adjacent segments
- They retain their original intersegmental arrangement in the thoracic and lumbar regions
- where they can be recognized as the *intercostal* and *lumbar* arteries

- **Umbilical Artery**

- Before the fusion of the two dorsal aortae, the umbilical arteries appear as continuations of their distal ends
- After fusion of the dorsal aortae, they appear as lateral branches of the single dorsal aorta
- each umbilical artery gets linked up with that **part of the fifth lumbar intersegmental artery** which forms the internal iliac artery
- part of the umbilical artery, between the aorta and the anastomosis with the internal iliac, disappears so that the umbilical artery is now seen as a branch of the internal iliac



- postnatal life, the proximal part of the umbilical artery becomes the *superior vesical artery*, while its distal part is obliterated to form the medial umbilical ligament
- S



- ① Blood enters right atrium from superior and inferior venae cavae.
- ② Blood in right atrium flows through right AV valve into right ventricle.
- ③ Contraction of right ventricle forces pulmonary valve open.
- ④ Blood flows through pulmonary valve into pulmonary trunk.
- ⑤ Blood is distributed by right and left pulmonary arteries to the lungs, where it unloads CO_2 and loads O_2 .
- ⑥ Blood returns from lungs via pulmonary veins to left atrium.
- ⑦ Blood in left atrium flows through left AV valve into left ventricle.
- ⑧ Contraction of left ventricle (simultaneous with step 3) forces aortic valve open.
- ⑨ Blood flows through aortic valve into ascending aorta.
- ⑩ Blood in aorta is distributed to every organ in the body, where it unloads O_2 and loads CO_2 .
- ⑪ Blood returns to heart via venae cavae.

• **Fetal Circulation**

- different from that of adult circulation
- 3 basic reasons for this difference:
 - 1. Blood in the fetus is oxygenated by placenta and not by lungs
 - 2. During fetal life, the lungs are collapsed, hence the resistance to blood flow through the lung is much higher
 - result, only minimal amount of blood passes through the lungs to supply oxygen and nutrients to the lungs.
- 3. Portal circulation is of little significance

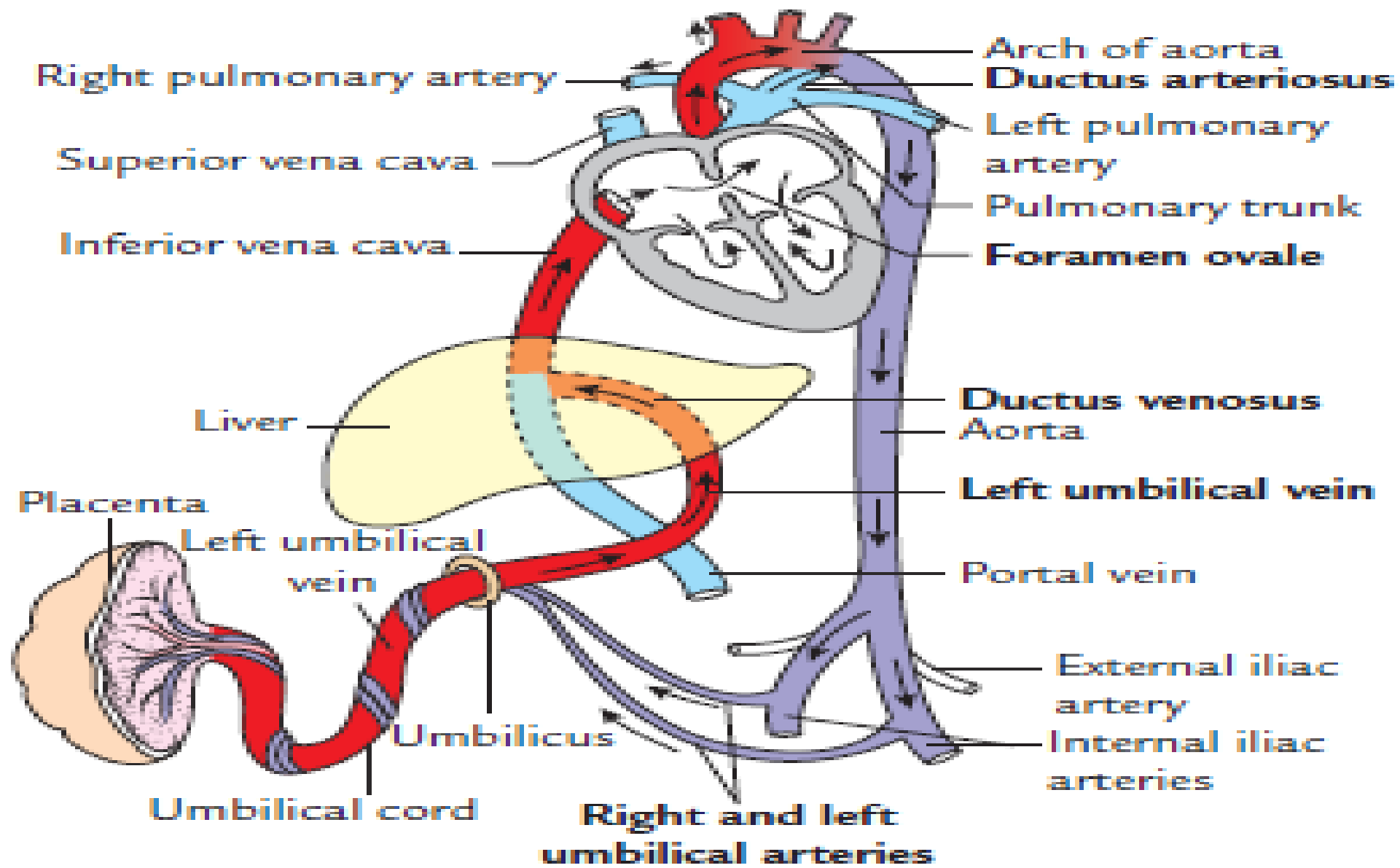
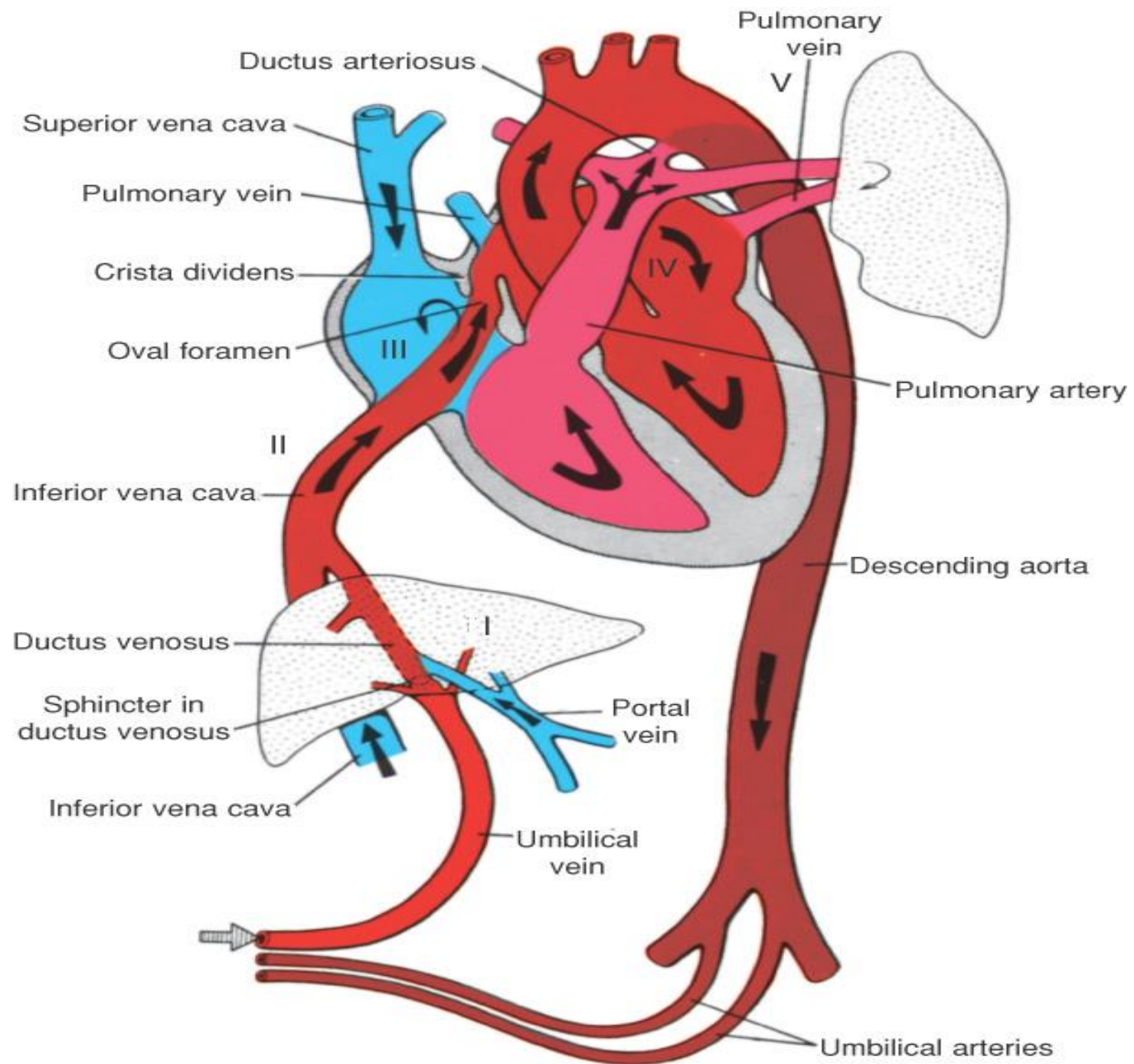


Fig. 19.25 Fetal circulation.

- Before birth, the oxygenated blood from the placenta returns to the fetus by the **left umbilical vein**
- umbilical vein traverses through the **umbilical cord**
- enters the abdomen through **umbilicus** & passes to the liver through falciform ligament
- In the liver, left umbilical vein joins the left branch of portal vein
- most of blood from left umbilical vein bypasses the sinusoids of the liver by passing through ***ductus venosus***—a channel that connects left branch of portal vein to the **inferior vena cava**
- small amount of blood of left umbilical vein enters the liver sinusoids & mixes with the blood from the portal circulation

- inferior vena cava opens into right atrium
- inferior vena cava, richly oxygenated blood mixes with deoxygenated blood returning from the lower limbs
- this blood enters in the right atrium, it is guided by the valve of inferior vena cava towards foramen ovale
- most of blood from the inferior vena cava passes through foramen ovale into the left atrium
- small amount of blood is prevented from entering left atrium by the lower edge of septum secundum—the *crista dividens*



- Some blood remains in the right atrium
- it mixes with the deoxygenated blood from the superior vena cava and passes to the right ventricle
- Blood from the right ventricle passes through pulmonary trunk, and right and left pulmonary arteries
- due to high resistance in pulmonary tissue during fetal life only a small amount of blood enters the **pulmonary circulation**
- most of it passes into the aorta through the **ductus arteriosus (DA)**—a channel that connects the left pulmonary artery to arch of aorta

- left atrium receives mainly rich oxygenated blood from the right atrium through the **foramen ovale**
- a small amount of deoxygenated blood enters the left atrium from the lungs through the pulmonary veins
- Blood from the left atrium passes to the left ventricle
- Blood from the left ventricle enters into the ascending aorta
- through its 3 large branches (brachiocephalic artery, left common carotid artery & left subclavian artery) supply oxygenated blood to head, neck, brain, and superior extremity

1 Blood bypasses the lungs by flowing from the pulmonary trunk through the ductus arteriosus to the aorta.

2 Blood also bypasses the lungs by flowing from the right to the left atrium through the foramen ovale.

3 Blood bypasses the liver sinusoids by flowing through the ductus venosus.

4 Oxygenated blood from the placenta is passed to the fetus by the umbilical vein.

5 Deoxygenated blood is carried from the fetus to the placenta through the umbilical arteries.

Superior vena cava

Aortic arch

Ductus arteriosus

Pulmonary trunk

Foramen ovale

Inferior vena cava

Liver

Ductus
venosus

Hepatic
portal vein

Umbilical vein

Fetal umbilicus

Umbilical cord

Umbilical arteries

Abdominal
aorta

Kidney

Common iliac
artery

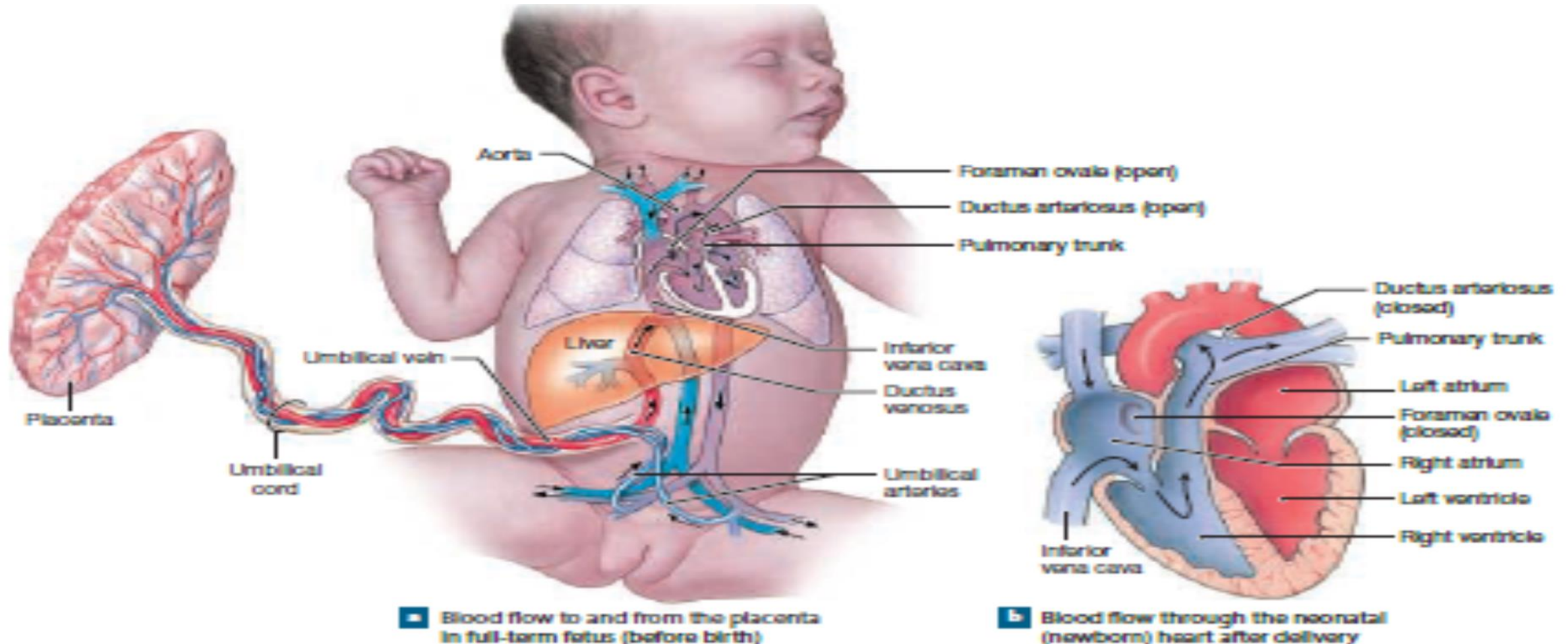
Internal iliac
arteries

- coronary and carotid arteries are the first branches of the aorta
- heart musculature and brain are supplied by well oxygenated blood
- arch of aorta receives poorly oxygenated blood from the pulmonary trunk through the ductus arteriosus
- blood is distributed by the aorta & common iliac arteries to the lower part of the body
- lower part of the body is supplied with relatively less oxygenated blood as compared with the upper part of the body
- **2 umbilical arteries** arising from internal iliac branches of common iliac arteries pass through umbilicus
- enter the placenta through the umbilical cord, where it is oxygenated

- **Changes in Fetal Circulation Just after Birth**
- After birth, placenta—**fetal organ of respiration**—is separated from newborn & lung starts oxygenating the blood
- **1. Umbilical vein** no longer carries any blood from the placenta, obliterates & forms a fibrous ligament called *ligamentum teres hepatis*
- **2. Ductus venosus** obliterates to form a fibrous ligament called *ligamentum venosum*
- **3.** As the lungs are inflated and the pulmonary circulation is established, pulmonary veins bring more blood to the left atrium

- pressure of blood in the left atrium is more than that in the right atrium the septum primum is pushed to the right & the foramen ovale is closed
- first the closure of foramen ovale is physiological
- later septum primum fuses with the septum secundum & there is an anatomical closure of foramen ovale
- closed **foramen ovale** forms *fossa ovalis*
- **4.** Ductus Arteriosus obliterates to form a fibrous ligament called the *ligamentum arteriosum*

- **5. Umbilical arteries** (right and left) obliterate
- their proximal parts remain open
- proximal parts of umbilical arteries form ***superior vesicle arteries***
- distal parts form fibrous ligaments called ***medial umbilical ligaments***



<i>Vessel</i>	<i>Remnant</i>	<i>Reasons for closure</i>
Umbilical arteries	Proximal part— superior vesical artery Distal part— fibrosed—medial umbilical ligament	Contraction of thick muscle wall
Left umbilical vein	Ligamentum teres of the liver	
Ductus venosus	Ligamentum venosum	• Functional closure within minutes after birth • Structural closure occurs within 3–7 days • Closure of ductus venosus is caused by strong contraction of its muscle wall
Ductus arteriosus	Ligamentum arteriosum	Smooth muscle contraction At birth, opposite direction of blood flow from aorta to pulmonary artery supplies more oxygenated blood than before that causes smooth muscle contraction

- **Clinical Correlation**

- 1. ***Patent ductus arteriosus (PDA)*** :

- occurs when ductus arteriosus, a connecting channel between left pulmonary artery and the arch of aorta, fails to close
- The PDA causes shunting of blood from aorta back into the pulmonary circulation
- The PDA is one of the most common congenital anomalies of the great vessels
- occurring in 8/10,000 births, especially in premature female babies born to mothers who had suffered from *rubella infection* in the early part of pregnancy

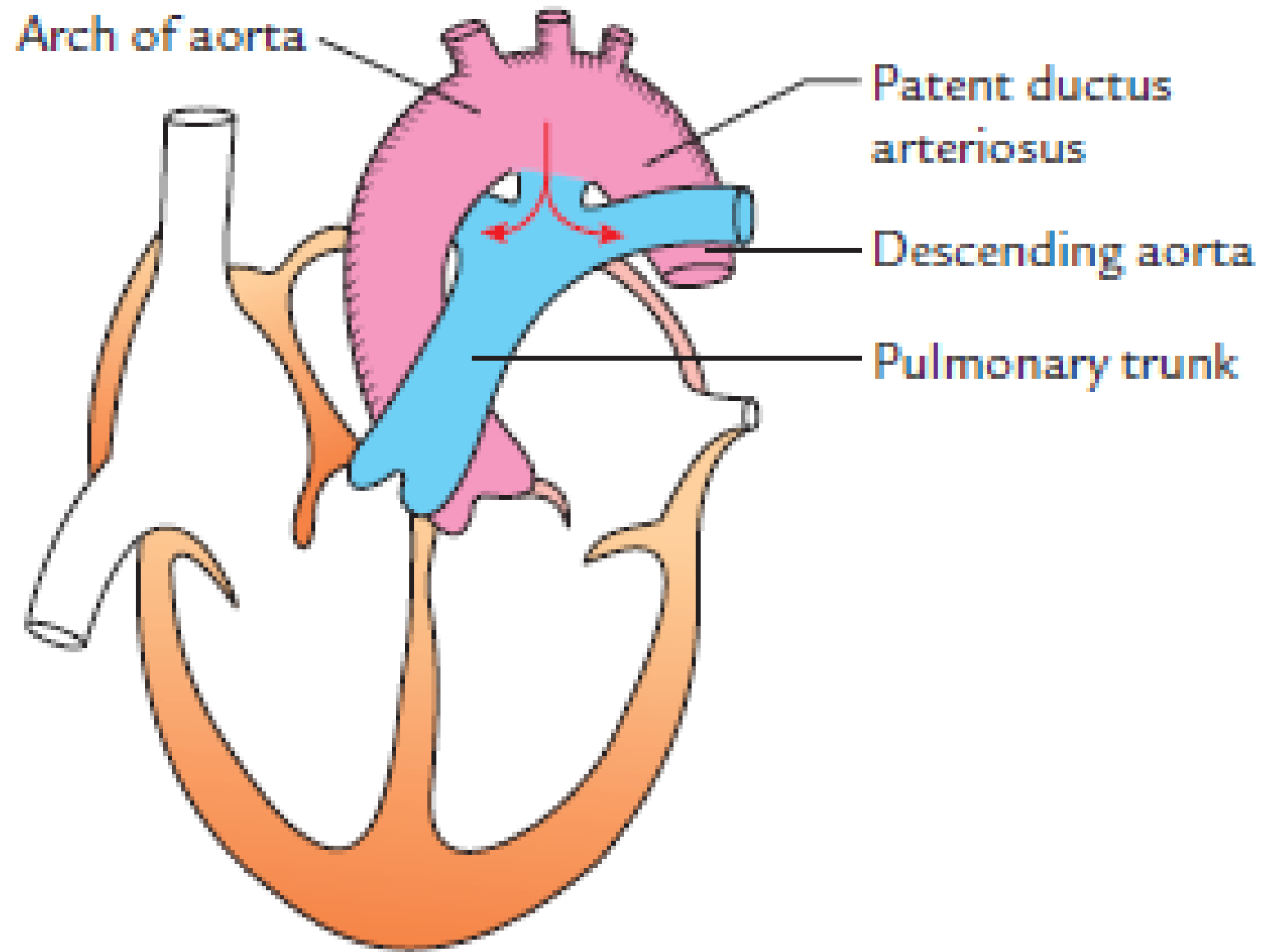


Fig. 19.8 Patent ductus arteriosus.

- **Functional closure** of the ductus arteriosus (DA) occurs at birth by contraction of smooth muscles of the DA.
- **Anatomical closure** occurs by proliferation of tunica intima of DA 1–3 months after birth
- It is mediated by **bradykinin**—a substance released from lungs, during their initial inflation

- **2. *Coarctation of aorta* :**

- The coarctation means narrowing of the aorta
- It occurs due to extension of the process of obliteration of DA into the aorta
- The coarctation of aorta is of two types: *preductal* and *postductal*
- **(a) Preductal type of coarctation of aorta:**
 - In this type, a narrow segment of arch of aorta is proximal to entrance of the DA
 - The DA persists in this type.
- **(b) Postductal type of coarctation of aorta:**
 - In this type, a narrow segment of arch of aorta is distal to the entrance of the DA
 - The DA usually obliterates in this type

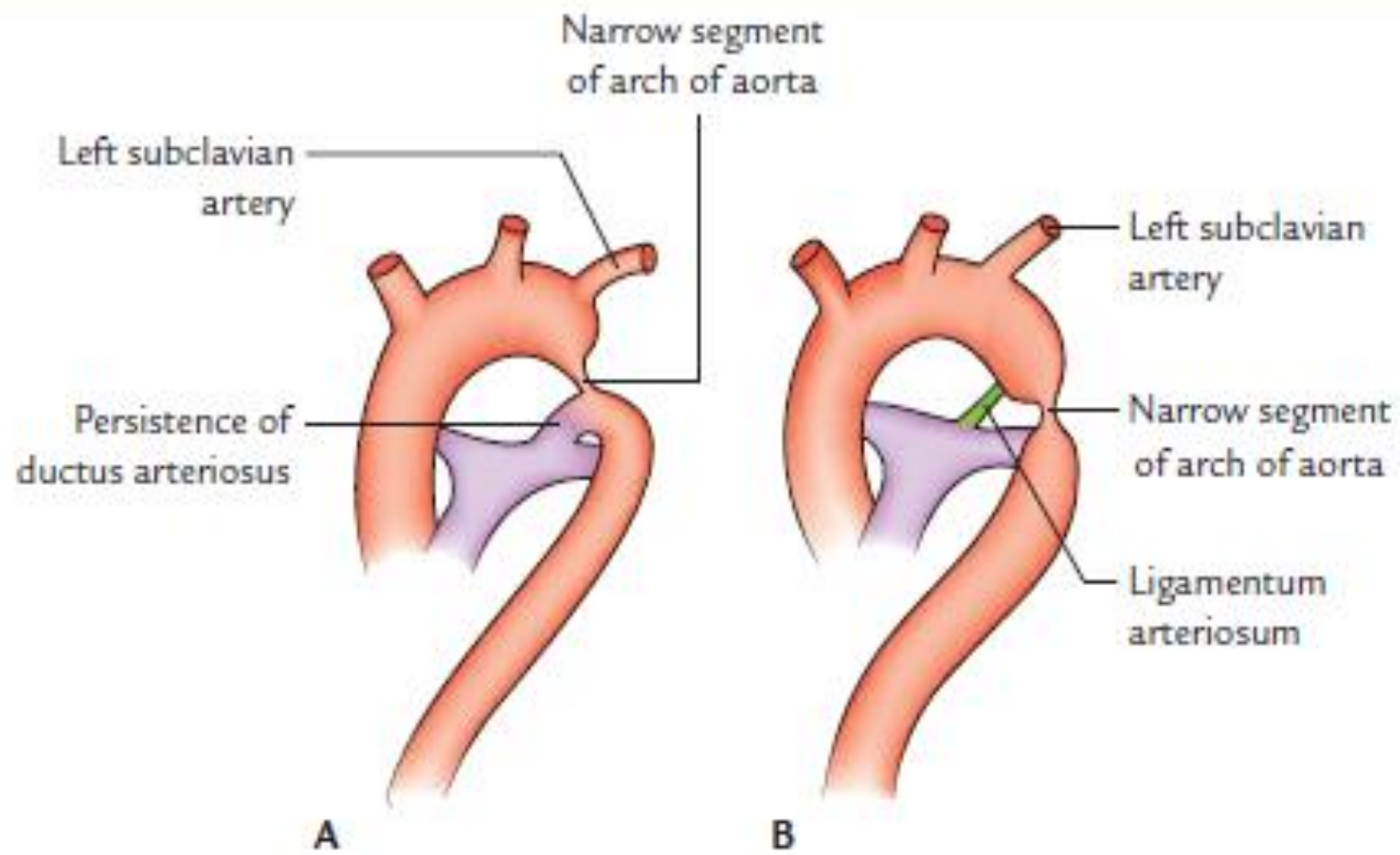


Fig. 19.9 Coarctation of aorta.